

Unit II: Inferential Statistics & Hypothesis Testing

CSE552 — Business Analytics

Session 2026-27

Learning Outcomes

By the end of this unit, you will be able to:

- **Population:** Start with the entire group you want to study.
- **Samples:** Learn how to pick a small, random part of that group.
- **Sampling Distributions:** See how sample results spread out when you take many samples.
- **Normal Distribution:** Understand the bell-shaped curve data often follows.
- **Central Limit Theorem (CLT):** Learn the rule saying sample means form a normal curve if the sample is large.
- **Central Limit Theorem - Demonstration:** Look at visual or computer examples of the rule working.
- **Application of CLT:** See how to use the rule for real-world estimates.
- **Z-Score:** Learn how to measure distance from the mean in standard units.
- **P-Value:** Understand the probability of getting your test results.
- **Rejection region:** Find the cutoff zone where you decide things are unusual.
- **Hypothesis testing:** Finish by testing claims about your data

1 Normal Distribution

Demonstration : <https://www.mathsisfun.com/data/standard-normal-distribution-table.html>

1.1 Normal Distribution

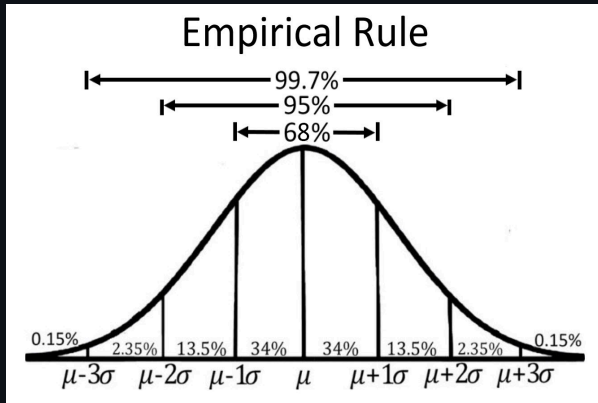
Definition

The **Normal (Gaussian) Distribution** is a continuous probability distribution characterized by its symmetric, bell-shaped curve.

A Normal Distribution is a bell-shaped, symmetric probability distribution where most observations are concentrated around the mean, and fewer observations occur as we move away from the mean.

Characteristics

-  Bell-shaped curve
-  Symmetric about the mean
-  Mean = Median = Mode
-  Total area under the curve = 1 (100%)
-  Defined by:
 - Mean (μ) – Center of the distribution
 - Standard Deviation (σ) – Spread of the data



The 68–95–99.7 Rule

Approximately:

- 68% of observations lie within $\pm 1\sigma$ of the mean.
- 95% lie within $\pm 2\sigma$.
- 99.7% lie within $\pm 3\sigma$.

2. Central Limit Theorem (CLT)

Statement

CLT: For a sufficiently large sample size ($n \geq 30$), the sampling distribution of the sample mean will be approximately **normally distributed**, regardless of the population's distribution.

Why is CLT Revolutionary?

- Before CLT: We could only work with normally distributed populations
- After CLT: We can apply normal distribution methods to ANY population
 - This is the foundation of almost all statistical inference!
- The Central Limit Theorem (CLT) works in practice by allowing you to make accurate predictions about a large population using small samples, even if the population data is skewed, uniform, or completely irregular.

How It Works (Step-by-Step)

- **Start with any population:** Imagine a population with a highly skewed distribution, like annual household incomes.
- **Take random samples:** Pull a random group of 30 people and calculate their average income.
- **Repeat the process:** Repeat this sampling process hundreds or thousands of times.
- **Plot the averages:** Create a histogram of only the sample averages you calculated.
- **The magic happens:** The resulting plot will form a perfectly symmetrical, normal bell curve.

Why This Matters to Statisticians

- **No population knowledge needed:** You do not need to know the shape of the entire population to study it.
- **Sample size benchmark:** A sample size of $n \geq 30$ is the general rule of thumb for the CLT to take effect.
- **Bridges sample to population:** It gives researchers the mathematical justification to use sample data to calculate margins of error and confidence intervals for entire nations.

3. Population vs. Sample

Definitions

Term	Definition	Symbol	Example
Population	Entire group of interest	N	All customers of a bank
Sample	Subset of the population	n	500 randomly selected customers
Parameter	Population characteristic	μ, σ	True average account balance
Statistic	Sample characteristic	$\bar{x}, s$	Sample average account balance

Why Sample?

- **Cost:** Studying entire population is expensive
- **Time:** Faster to collect sample data
- **Feasibility:** Some populations are infinite/inaccessible
- **Destruction:** Testing may destroy items

3.1 Sampling Methods

Probability Sampling (Random)

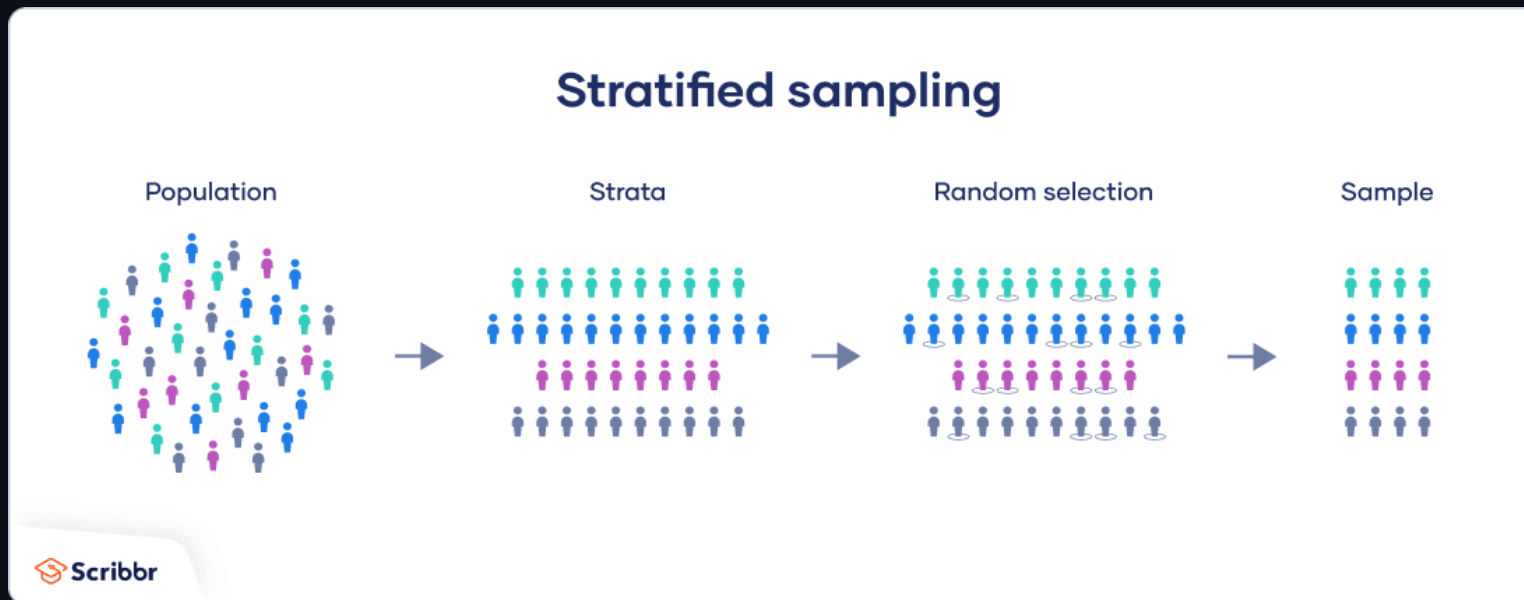
- | | |
|---------------------------|---|
| 1. Simple Random Sampling | → Every member has equal chance |
| 2. Stratified Sampling | → Divide into strata, non-overlapping subgroups[using a clear trait like age, job, or grade.] |
| 3. Systematic Sampling | → Select every k-th member |
| 4. Cluster Sampling | → Randomly select groups, survey all in group [Naturally occurring or geographic unit] |

Non-Probability Sampling

- | | |
|-------------------------|--|
| 1. Convenience Sampling | → Easy-to-reach members |
| 2. Judgmental Sampling | → Expert selects |
| 3. Snowball Sampling | → Referrals from existing participants |

A. Probability Sampling (Random - Every member has known chance)

Method	How it works	Best for
Simple Random	Randomly pick n from N	Homogeneous populations
Stratified	Divide into groups (strata), sample from each	Heterogeneous groups
Systematic	Pick every k-th member	Ordered lists
Cluster	Randomly pick groups, survey all in group	Geographically spread



Stratified Sampling vs Cluster Sampling

Stratified Sampling: National Education Survey	Cluster Sampling: Vaccine Distribution Study
Goal: Study academic performance across the country while ensuring minor groups are fairly represented.	Goal: Assess local health needs quickly without traveling to every household in a vast region.
Groups (Strata): Urban, suburban, and rural school districts.	Groups (Clusters): Cities or neighborhoods.
Action: Researchers pull a small, random sample of students from every single district type nationwide.	Action: Researchers randomly pick five specific cities out of fifty, then survey every single resident living inside those five selected cities.

Business Tip: Stratified sampling ensures representation across customer segments.

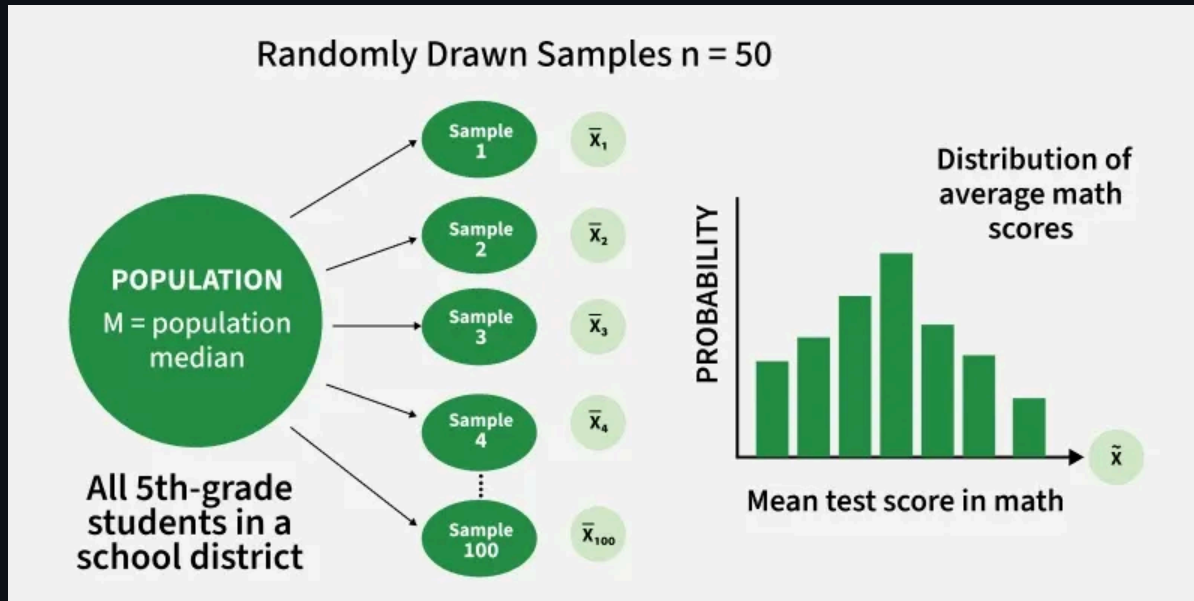
B. Non-Probability Sampling (Not random - Convenience-based)

Method	How it works	Risk
Convenience	Pick easiest to reach	Bias likely
Judgmental	Expert selects	Expert bias
Snowball	Referrals from existing	Limited reach

4. Sampling Distributions

Definition

A **sampling distribution** is the probability distribution of a statistic (e.g., sample mean) obtained from repeated samples of the same size from a population. It is also known as finite distribution.



Demonstration: https://onlinestatbook.com/stat_sim/sampling_dist/

Key Insight

- Individual observations vary
- But sample means tend to cluster around the population mean
- The spread of sample means is smaller than the spread of individual values

Main factors influencing the variability of a sampling distribution are:

1. Number Observed in a Population: The symbol for this variable is "N." It is the measure of observed activity in a given group of data.
2. Number Observed in Sample: The symbol for this variable is "n." It is the measure of observed activity in a random sample of data that is part of the larger grouping.
3. Method of Choosing Sample: How you chose the samples can account for variability in some cases.

4. Sampling Distributions

Reference: Penn State University — Applied Statistics

<https://online.stat.psu.edu/stat500/Lesson04>

4.1 Overview

In inferential statistics, we use **sample statistics** to estimate **population parameters**.

- **Statistic:** A characteristic of the sample (e.g., sample mean $\bar{x}$)
- **Parameter:** A characteristic of the population (e.g., population mean μ)

If we calculate a sample statistic from a random sample, that statistic is a **random variable**.

Sampling Distribution: The probability distribution of a statistic based on a large number of samples of size n from a given population.

Topic Objectives

Upon completion of this lesson, you should be able to:

1. Understand the meaning of **sampling distribution**
2. Apply the **Central Limit Theorem** to calculate approximate probabilities for sample means and sample proportions
3. Describe the sampling distribution of the **sample mean** and **sample proportion**
4. Identify situations in which the **normal distribution** and **t-distribution** may be used to approximate a sampling distribution

4.2 Sampling Distribution of the Sample Mean

Example 4.1: Pumpkin Weights

Population: Six pumpkins with weights (in pounds):

Pumpkin	A	B	C	D	E	F
Weight	19	14	15	9	10	17

Population Mean:

$$\mu = \frac{19 + 14 + 15 + 9 + 10 + 17}{6} = 14 \text{ pounds}$$

All Possible Samples (n = 2) :

Sampling without replacement, there are 15 possible samples:

Sample	Weights	$\bar{x}$	Probability
A, B	19, 14	16.5	1/15
A, C	19, 15	17.0	1/15
A, D	19, 9	14.0	1/15
A, E	19, 10	14.5	1/15
A, F	19, 17	18.0	1/15
B, C	14, 15	14.5	1/15
B, D	14, 9	11.5	1/15
B, E	14, 10	12.0	1/15
B, F	14, 17	15.5	1/15
C, D	15, 9	12.0	1/15
C, E	15, 10	12.5	1/15
C, F	15, 17	16.0	1/15
D, E	9, 10	9.5	1/15
D, F	9, 17	13.0	1/15
E, F	10, 17	13.5	1/15

Sampling Distribution (n = 2)

$\bar{x}$	9.5	11.5	12.0	12.5	13.0	13.5	14.0	14.5	15.5	16.0	16.5	17.0	18.0
P($\bar{x}$)	1/15	1/15	2/15	1/15	1/15	1/15	1/15	2/15	1/15	1/15	1/15	1/15	1/15

Mean of the Sample Means:

$$\mu_{\bar{x}} = \sum \bar{x}_i \cdot f(\bar{x}_i) = 14$$

The expected value of the sample mean equals the **population mean**.

Sampling Distribution (n = 5)

Sample	Weights	$\bar{x}$	Probability
A,B,C,D,E	19,14,15,9,10	13.4	1/6
A,B,C,D,F	19,14,15,9,17	14.8	1/6
A,B,C,E,F	19,14,15,10,17	15.0	1/6
A,B,D,E,F	19,14,9,10,17	13.8	1/6
A,C,D,E,F	19,15,9,10,17	14.0	1/6
B,C,D,E,F	14,15,9,10,17	13.0	1/6

$$\mu_{\bar{x}} = \frac{1}{6}(13 + 13.4 + 13.8 + 14.0 + 14.8 + 15.0) = 14$$

As sample size increases, sample means cluster more closely around μ .

Sampling Error

Sampling Error: The error resulting from using a sample characteristic to estimate a population characteristic.

Key Insight:

- The possible sample means cluster more closely around the population mean as n increases
- Thus, the **possible sampling error decreases as sample size increases**

When the Population is Normal

If the population is normally distributed with mean μ and standard deviation σ :

- The sampling distribution of $\bar{x}$ is also **normally distributed** for **any** sample size
- Mean: $\mu_{\bar{x}} = \mu$
- Standard deviation (Standard Error): $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$

Z-Score for Sample Mean:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Example 4.3: Speedboat Engines

Ford speedboat engines: $\mu = 220$ HP, $\sigma = 15$ HP, normally distributed.

Consumer Reports will dispute the claim if $\bar{x} < 215$ for $n = 4$.

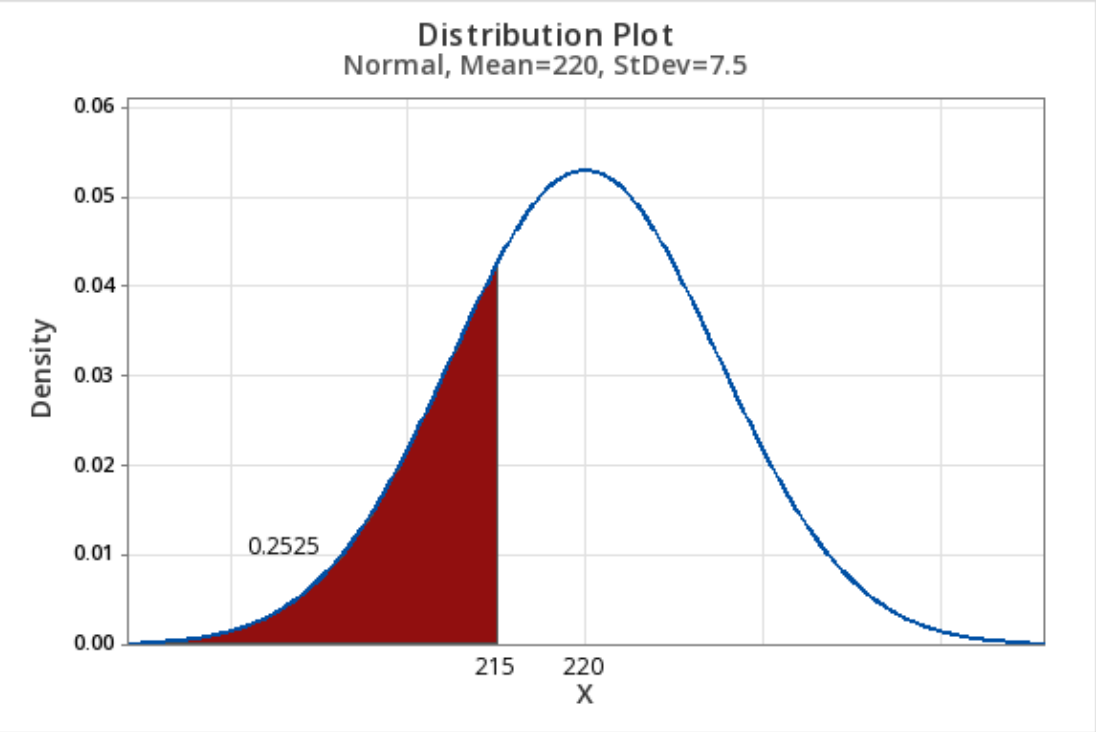
Solution:

$$SE = \frac{15}{\sqrt{4}} = 7.5$$

$$P(\bar{X} < 215) = P\left(Z < \frac{215 - 220}{7.5}\right) = P(Z < -0.67) \approx 0.2514$$

There is a **25.14%** chance the sample mean is less than 215 HP.

$$\text{As, } z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$



Number in the table represents $P(Z \leq z)$

z 0

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-3.6	.0002	.0002	.0001	.0001	.0001	.0001	.0001	.0001	.0001	.0001
-3.5	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002	.0002
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Try It! (n = 100)

If Consumer Reports samples 100 engines, what is $P(\bar{X} < 215)$?

$$SE = \frac{15}{\sqrt{100}} = 1.5$$

$$P(\bar{X} < 215) = P\left(Z < \frac{215 - 220}{1.5}\right) = P\left(Z < -\frac{10}{3}\right) \approx 0.00043$$

With $n = 100$, the probability drops to 0.043% — much less likely!

$$\text{As, } z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

4.4 When the Population is NOT Normal

This is where the **Central Limit Theorem (CLT)** comes in.

Central Limit Theorem: For a large sample size, $\bar{x}$ is approximately normally distributed, regardless of the distribution of the population.

- Mean: $\mu_{\bar{x}} = \mu$
- Standard Error: $\frac{\sigma}{\sqrt{n}}$

For this course: $n > 30$ is considered a large sample.

4.5 CLT Demonstration Notes

- If the population is skewed and sample size is small, $\bar{x}$ won't be normal
- If the population is normal, $\bar{x}$ looks normal even for $n = 2$
- If the population is skewed, $\bar{x}$ looks more normal as n gets larger
- In all cases, the mean of $\bar{x}$ is close to μ and $SE \approx \frac{\sigma}{\sqrt{n}}$

4.6 Sampling Distribution of $\bar{x}$ - Summary

The sampling distribution of the sample mean will have:

- **Mean:** $\mu_{\bar{x}} = \mu$
- **Standard Error:** $\frac{\sigma}{\sqrt{n}}$

It will be Normal (or approximately Normal) if:

1. ☒ The population distribution is Normal, **OR**
2. ☒ The sample size is large ($n > 30$)

Example 4.7: Weights of Baby Giraffes

Baby giraffe weights: $\mu = 125$ lbs, $\sigma = 15$ lbs. Sample of $n = 40$.

Q1. Probability that $\bar{X}$ is between 120 and 130:

$$SE = \frac{15}{\sqrt{40}}$$

$$P(120 < \bar{X} < 130) = P(-2.108 < Z < 2.108) = 0.9826 - 0.0174 = \mathbf{0.9652}$$

The probability that the sample mean of the 40 giraffes is between 120 and 130 lbs is 96.52%.

Consider formula: $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$;

Note the values from the Z-table for positive values given in the slide next to this question.

Q2. 75th percentile of sample means:

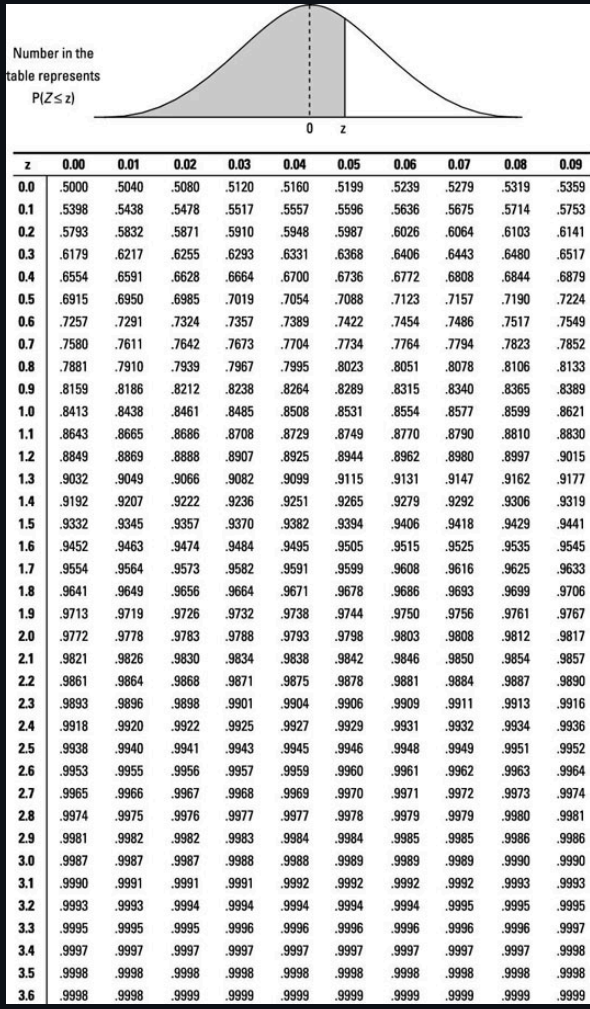
$$P(Z < \alpha) = 0.75$$

Using the Z-table $\alpha = 0.6745$

Consider formula: $z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$; $z = \alpha$

$$\bar{X} = 0.6745 \cdot \frac{15}{\sqrt{40}} + 125 \approx \mathbf{126.6 \text{ lbs}}$$

The 75th percentile of all the sample means of size is pounds.



Sample Proportion

A sample proportion is a statistic that measures the fraction of successes in a specific group or sample. It is written as $\hat{p}$ (read as "p-hat"), and it helps estimate the true proportion of a whole population.

Formula: $(\hat{p} = \frac{x}{n})$, where

- x is the number of successful outcomes and
- n is the total sample size.

Example: If 50 out of 200 surveyed people like blue, the sample proportion is $(\frac{50}{200} = 0.25)$ (or 25%)

Sampling Distribution of the Sample Proportion

Key Notation

Symbol	Meaning
p	Population proportion (fixed)
n	Sample size
$\hat{p}$	Sample proportion (varies by sample)

Example 4.5: Favorite Color

Family of 5 children: 2 prefer Blue ($p = 0.40$).

All samples of $n = 2$ (without replacement):

$\hat{p}$	0	1/2	1
Probability	3/10	6/10	1/10

Name	Alex (A)	Betina (B)	Carly (C)	Debbie (D)	Edward (E)
Color	Green	Blue	Yellow	Purple	Blue

Sample	P(Blue)	Probability
AB	1/2	1/10
AC	0	1/10
AD	0	1/10
AE	1/2	1/10
BC	1/2	1/10
BD	1/2	1/10
BE	1	1/10
CD	0	1/10
CE	1/2	1/10
DE	1/2	1/10

The true proportion $p = 0.40$ may not even be achievable with small n .

Sample	P(Blue)	Probability
AB	1/2	1/10
AC	0	1/10
AD	0	1/10
AE	1/2	1/10
BC	1/2	1/10
BD	1/2	1/10
BE	1	1/10
CD	0	1/10
CE	1/2	1/10
DE	1/2	1/10

The probability mass function (PMF) is: (2 prefer Blue, 3 prefer other colors)

P(Blue) $\hat{p}$	0	1/2	1
Probability	03/10	06/10	1/10

💡 Key Insight: Notice that the true population proportion is 0.40, but with a sample of just 2, we can never actually get $\hat{p} = 0.40$

! Our sample proportion is "stuck" at 0, 0.5, or 1. This is why small samples can be misleading.

Normal Distribution: If the sample size is large enough, the distribution of the sample proportion looks like a normal bell curve.

Normal Approximation to the Binomial

The sample proportion can be viewed as a **mean** of Bernoulli trials:

$$\hat{p} = \frac{\sum_{i=1}^n Y_i}{n} = \frac{X}{n}$$

Since $\hat{p}$ is a mean, we can apply the CLT for large samples!

Mean of $\hat{p}$: $\mu_{\hat{p}} = p$

Standard Error of $\hat{p}$: $\sqrt{\frac{p(1-p)}{n}}$

Conditions for Normal Approximation

Different textbooks use different thresholds:

Era	Condition 1	Condition 2
Book (Mini tab)	$np \geq 5$	$n(1 - p) \geq 5$
1990s–2000s	$np \geq 10$	$n(1 - p) \geq 10$
Current	$np \geq 15$	$n(1 - p) \geq 15$

If satisfied, $\hat{p}$ is approximately Normal with mean p and SE $\sqrt{\frac{p(1-p)}{n}}$.

Z-Score for Sample Proportion

$$z = \frac{\hat{p} - p}{\sqrt{\frac{p(1-p)}{n}}}$$

Example 4.7: iPhone Users

Known: $p = 0.43$ of Americans own an iPhone. Sample $n = 50$.

Q: What is the probability that the proportion of the sample who owned an iPhone is between 45% and 50%?

Check conditions:

- $np = 50(0.43) = 21.5 \geq 5$ ✓
- $n(1 - p) = 50(0.57) = 28.5 \geq 5$ ✓

Find $P(0.45 < \hat{p} < 0.50)$:

$$SE = \sqrt{\frac{0.43(0.57)}{50}} \approx 0.07$$

$$P(0.45 < \hat{p} < 0.50) = P(0.286 < Z < 1) = 0.8413 - 0.6126 = \mathbf{0.2287}$$

There is a **22.87%** chance of seeing a sample proportion between 45% and 50%.

Try It! ($n = 75$)

Sample $n = 75$. Find $P(\hat{p} > 0.50)$.

Check conditions:

- $np = 75(0.43) = 32.25 \geq 5$ ✓
- $n(1 - p) = 75(0.57) = 42.75 \geq 5$ ✓

$$SE = \sqrt{\frac{0.43(0.57)}{75}} \approx 0.057$$

$$P(\hat{p} > 0.50) = P(Z > 1.22) = 1 - 0.8888 = \mathbf{0.1112}$$

There is an 11.1% chance of getting a sample proportion of 50% or higher.

Lesson Summary

Concept	Sample Mean $\bar{x}$	Sample Proportion $\hat{p}$
Mean	μ	p
Standard Error	$\frac{\sigma}{\sqrt{n}}$	$\sqrt{\frac{p(1-p)}{n}}$
Normal if...	Population Normal or $n > 30$	$np \geq 5$ and $n(1 - p) \geq 5$
Z-Score	$\frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$	$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}}$

The Central Limit Theorem is the bridge from probability to **statistical inference**.

5 Hypothesis Testing

Reference:

<https://online.stat.psu.edu/statprogram/reviews/statistical-concepts/hypothesis-testing>

Overview

Hypothesis testing is a fundamental procedure in statistics used to make decisions about population parameters based on sample data.

Key Idea: We start with an assumption (the null hypothesis) and use data to determine whether there is sufficient evidence to reject that assumption.

This review covers:

- The criminal trial analogy
- Type I and Type II errors
- The critical value approach
- The P-value approach
- Worked examples for one-sample mean and one-sample proportion

The Criminal Trial Analogy

Hypothesis testing mirrors the U.S. criminal justice system:

Trial Concept	Statistical Equivalent
Defendant is innocent until proven guilty	We assume the null hypothesis is true
Prosecution collects evidence	We collect sample data
Jury decides: guilty or not guilty	We decide: reject or fail to reject H_0

Hypotheses in the Analogy

- H_0 : Defendant is **not guilty** (innocent)
- H_A : Defendant is **guilty**

We never "accept" the null hypothesis — we only **fail to reject** it when evidence is insufficient.

Setting Up Hypotheses

For a population mean μ , there are three possible test types:

Type	Null Hypothesis	Alternative Hypothesis	When to Use
Right-tailed	$H_0 : \mu = \mu_0$	$H_A : \mu > \mu_0$	Want to show mean is greater than a value
Left-tailed	$H_0 : \mu = \mu_0$	$H_A : \mu < \mu_0$	Want to show mean is less than a value
Two-tailed	$H_0 : \mu = \mu_0$	$H_A : \mu \neq \mu_0$	Want to show mean differs from a value

The **alternative hypothesis** (H_A) is what the researcher wants to demonstrate.

Type I and Type II Errors

Whenever we conduct a hypothesis test, two types of errors can occur:

	H_0 is True	H_0 is False
Reject H_0	Type I Error (α)	Correct Decision (Power)
Fail to Reject H_0	Correct Decision	Type II Error (β)

In the Trial Analogy

- **Type I Error:** Finding an innocent person guilty
- **Type II Error:** Finding a guilty person not guilty

α = significance level = probability of Type I error. Common values: 0.01, 0.05, 0.10.

Approach 1: Critical Value Approach

Determines "likely" or "unlikely" by comparing the test statistic to a **cutoff value**.

Four Steps:

1. **Specify** the null and alternative hypotheses
2. **Calculate** the test statistic (assuming H_0 is true)
3. **Determine** the critical value based on α (significance level)
4. **Compare** the test statistic to the critical value

| If the test statistic is **more extreme** than the critical value → **Reject H_0**

Test Statistic for Population Mean

When σ is unknown (most common case), we use the **t-statistic**:

$$t^* = \frac{\bar{x} - \mu_0}{s / \sqrt{n}}$$

This follows a **t-distribution** with $n - 1$ degrees of freedom.

	Test Statistic (t^*)	Critical Value (t_{α})
Where it comes from	Your sample data	A table (or software)
How you get it	Calculate using the formula	Look up using α and df
Uses your data?	✔ Yes ($\bar{x}$, s , n)	✗ No
Uses μ ?	✔ Yes (μ_0 in the formula)	✗ No

Critical Values by Tail Type

Test Type	Rejection Region
Right-tailed	Reject if $t^* > t_{\alpha,n-1}$
Left-tailed	Reject if $t^* < -t_{\alpha,n-1}$
Two-tailed	Reject if $t^* < -t_{\alpha/2,n-1}$ or $t^* > t_{\alpha/2,n-1}$

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

A researcher wants to test if the mean GPA of math majors is greater than 3.0. She collects data from a random sample of math majors.

Symbol	What it means	Value
μ_0	The hypothesized population mean we're testing against	3.0
$\bar{x}$	The sample mean from our actual data	3.25
s	The sample standard deviation	0.40
n	The sample size	15
α	The significance level	0.05

Critical Value Example: (a) Right-Tailed Test

Scenario: Test if mean GPA of math majors $\mu > 3$; at 5 level of significance.

 Step 1: State the Hypotheses: - $H_0 : \mu = 3$; $H_A : \mu > 3$ [Right-tailed test because H_a has the ">" sign]

 Step 2: Calculate the Test Statistic (t^*): $t^* = (\bar{x} - \mu_0) / (s / \sqrt{n})$

$$t^* = (3.25 - 3.0) / (0.40 / \sqrt{15}) = 0.25 / (0.40 / 3.873) = 0.25 / 0.1033 = 2.42$$

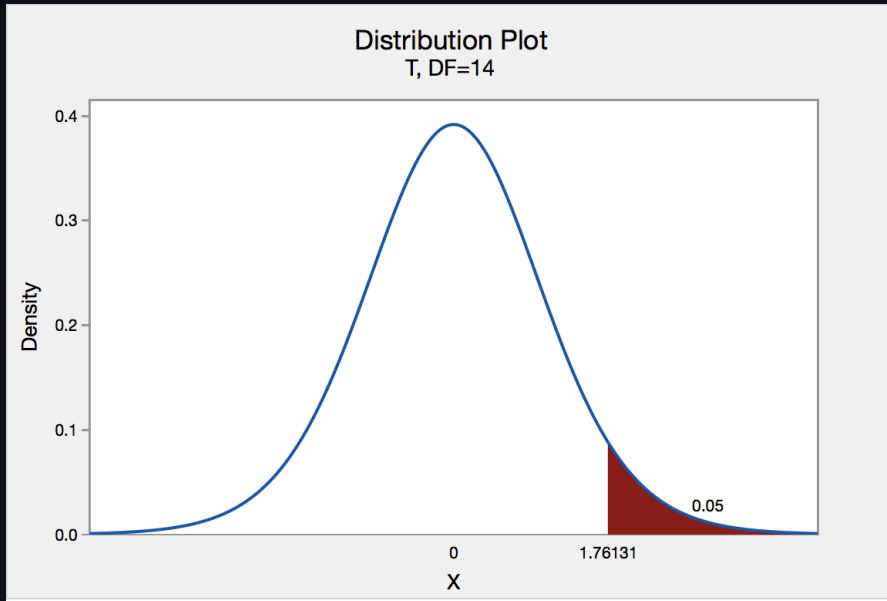
 Step 3: Find the Critical Value (t_{α})

From the t-distribution table with: $\alpha = 0.05$ (one-tailed, right side) AND $df = n - 1 = 15 - 1 = 14$

$$t_{0.05,14} = 1.7613$$

 Step 4: Compare t^* vs. t_{α} : Test Statistic ($t^* = 2.42$) AND Critical Value ($t_{\alpha} = 1.76$)

$t^* > t_{\alpha}$; $\rightarrow$ REJECT H_0 [Our test statistic is more extreme than the critical value.]



$$t^* = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Reject H_0 if $t^* > 1.7613$

The rejection region is the area to the **right** of 1.7613 under the t_{14} curve.

At the $\alpha = 0.05$ significance level, there is sufficient evidence to conclude that the mean GPA of math majors is greater than 3.0.

Critical Value Example: (b) Left-Tailed Test

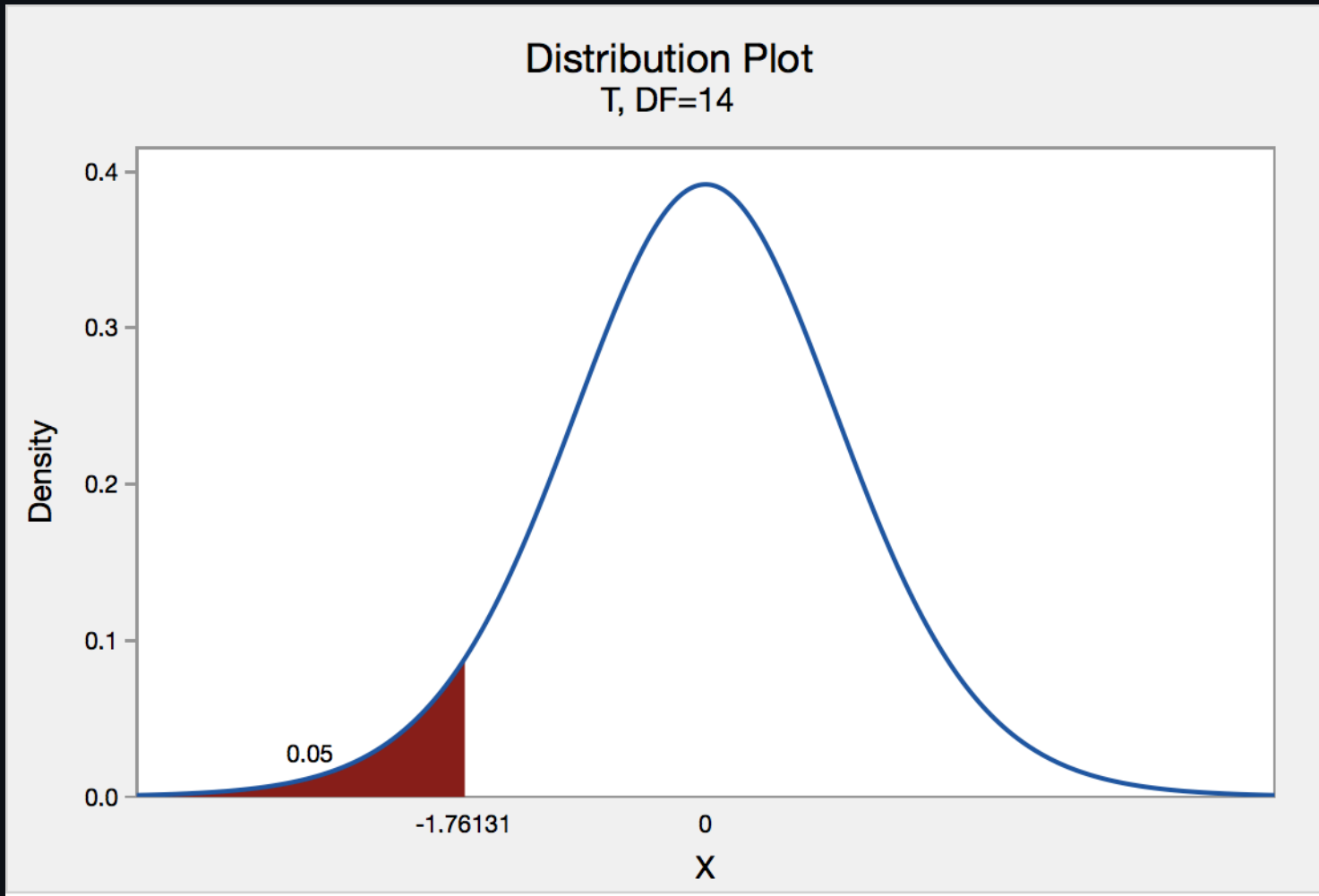
Scenario: Test if mean GPA $\mu < 3$

- $H_0 : \mu = 3$ vs. $H_A : \mu < 3$
- $n = 15, df = 14$
- $\alpha = 0.05$

Critical Value: $-t_{0.05,14} = -1.7613$

Reject H_0 if $t^* < -1.7613$

The rejection region is the area to the left of -1.7613 .



Critical Value Example: (c) Two-Tailed Test

Scenario: Test if mean GPA $\mu \neq 3$

- $H_0 : \mu = 3$ vs. $H_A : \mu \neq 3$
- $n = 15, df = 14$
- $\alpha = 0.05$, so $\alpha/2 = 0.025$

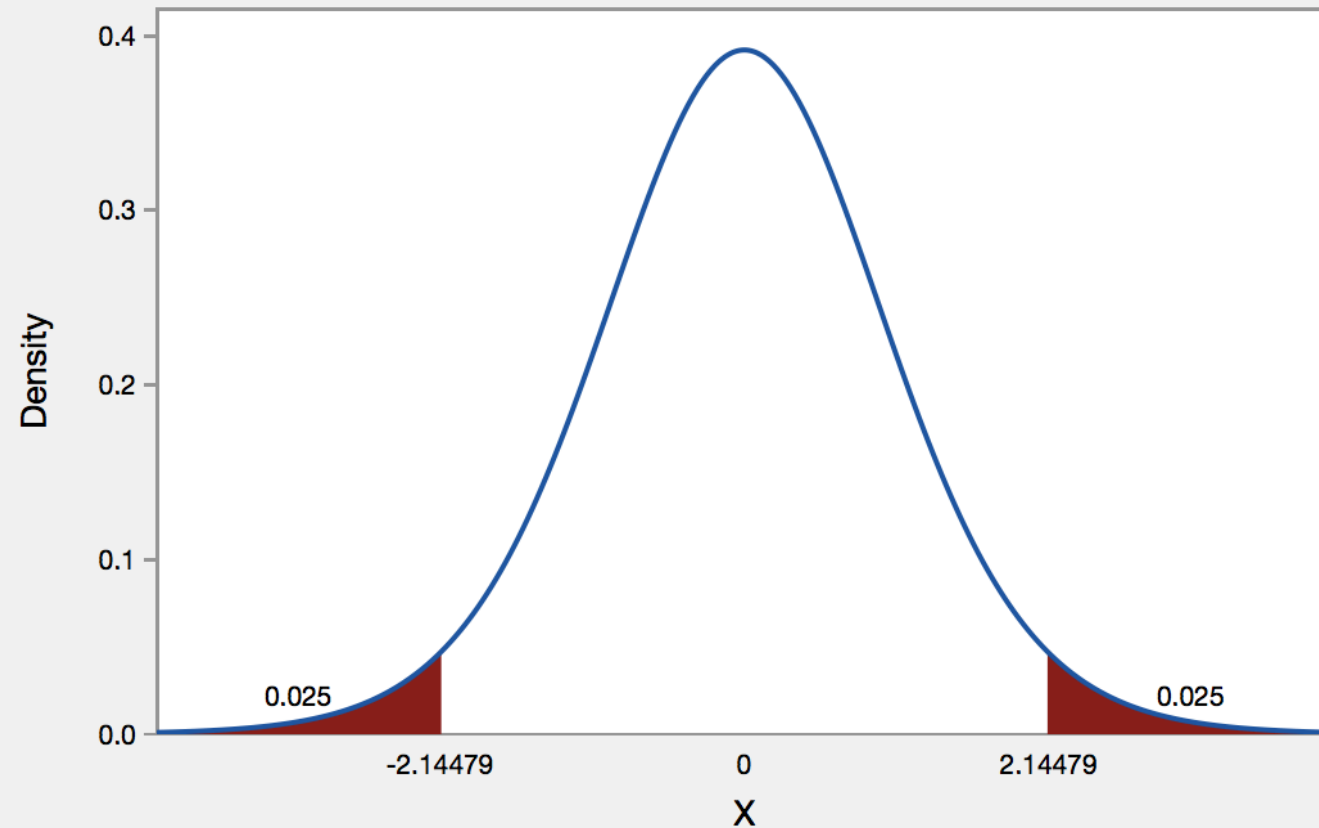
Critical Values:

- $-t_{0.025,14} = -2.1448$
- $t_{0.025,14} = 2.1448$

Reject H_0 if $t^* < -2.1448$ or $t^* > 2.1448$

Two rejection regions — one in each tail.

Distribution Plot
T, DF=14



Approach 2: P-Value Approach

Determines "likely" or "unlikely" by calculating a **probability**.

Definition: The P-value is the probability of getting a test statistic at least as extreme as the one we observed, assuming H_0 is true.

Four Steps:

1. **Specify** the null and alternative hypotheses
2. **Calculate** the test statistic (assuming H_0 is true)
3. **Calculate** the P-value: *"If H_0 is true, what's the probability of observing a test statistic at least as extreme as ours?"*
4. **Compare** the P-value to α

If $\text{P-value} \leq \alpha \rightarrow \text{Reject } H_0$ (evidence is "unlikely" under H_0)

P-Value by Tail Type

Test Type	P-Value Calculation
Right-tailed	$P(T \geq t^*)$
Left-tailed	$P(T \leq t^*)$
Two-tailed	$2 \times P(T \geq t^*)$

For a two-tailed test, the P-value is always twice the one-tailed P-value.

t Table											
cum. prob	t _{.50}	t _{.75}	t _{.80}	t _{.85}	t _{.90}	t _{.95}	t _{.975}	t _{.99}	t _{.995}	t _{.999}	t _{.9995}
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073

How to obtain the p value in this case: $P(T_{14} \geq 2.42)$

P-value is between 0.025 and 0.010. Rough estimate: $T_{14} \sim 0.015$ (close to 0.0148!)

t Table											
cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
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3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
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11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
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13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073

P-Value Example: (a) Right-Tailed Test

Q: Same scenario — testing if math majors' mean GPA is greater than 3.0 — but now using the P-value approach instead of the critical value approach.

Symbol	Meaning	Value
μ_0	Hypothesized mean	3.0
$\bar{x}$	Sample mean	3.25
s	Sample standard deviation	0.40
n	Sample size	15
α	Significance level	0.05

 Step 1: State the Hypotheses

$H_0 : \mu = 3.0$ **AND** $H_a : \mu > 3.0$; Right-tailed test because H_a has the ">" sign.

 Step 2: Calculate the Test Statistic (t^*): $t^* = (\bar{x} - \mu_0) / (s / \sqrt{n}) = (3.25 - 3.0) / (0.40 / \sqrt{15})$
 $t^* = 0.25 / 0.1033 = 2.42$

 Step 3: Calculate the P-Value: $P(T_{14} \geq 2.42)$; $P - value = 0.0148$

 Step 4: Compare P-Value to α :

| The P-value is smaller $\rightarrow$ REJECT H_0

 Step 5: Conclusion

- At the $\alpha = 0.05$ significance level, there is sufficient evidence to conclude that the mean GPA of math majors is greater than 3.0.
- (The P-value of 0.0148 tells us the evidence is fairly strong — we'd only see this result by random chance about 1.5% of the time if H_0 were true.)

| summary

Scenario: $n = 15$, $df = 14$, $\alpha = 0.05$, observed $t^* = 2.5$

P-value = $P(T_{14} > 2.5) = 0.0127$

Since $0.0127 \leq 0.05$:

| Reject H_0 — there is sufficient evidence that $\mu > 3$

The observed result would only happen 1.27% of the time if H_0 were true.

P-Value Example: (b) Two-Tailed Test

Scenario: Same setup, but $H_A : \mu \neq 3$, observed $t^* = -2.5$

$$\text{P-value} = 2 \times P(T_{14} > 2.5) = 2 \times 0.0127 = 0.0254$$

Since $0.0254 \leq 0.05$:

■ Reject H_0 — there is sufficient evidence that $\mu \neq 3$

Example 1: Brinell Hardness Scores

An engineer tests whether the mean Brinell hardness of ductile iron exceeds 170.

- $H_0 : \mu = 170$ vs. $H_A : \mu > 170$
- $n = 25, \bar{x} = 172.52, s = 10.31$
- $SE = 10.31 / \sqrt{25} = 2.06$
- $t^* = (172.52 - 170) / 2.06 = 1.22$
- P-value = 0.117

Critical Value Approach

Critical value $t_{0.05,24} = 1.7109$; Since $1.22 < 1.7109 \rightarrow$ Fail to reject H_0

P-Value Approach

Since $0.117 > 0.05 \rightarrow$ Fail to reject H_0 ; **Conclusion:** Insufficient evidence that mean hardness exceeds 170.

Example 2: Height of Sunflowers

A biologist tests whether sunflower seedlings are shorter than 15.7 cm.

- $H_0 : \mu = 15.7$ vs. $H_A : \mu < 15.7$
- $n = 33, \bar{x} = 13.664, s = 2.544$
- $SE = 2.544/\sqrt{33} = 0.443$
- $t^* = (13.664 - 15.7)/0.443 = -4.60$
- P-value ≈ 0.000

Critical Value Approach

Critical value $-t_{0.05,32} = -1.6939$

Since $-4.60 < -1.6939 \rightarrow$ **Reject H_0**

P-Value Approach

Since P-value $< 0.001 < 0.05 \rightarrow$ **Reject H_0**

Conclusion: Strong evidence that mean height is less than 15.7 cm.

Example 3: Gum Thickness

A quality specialist tests whether gum thickness differs from 7.5 (two-tailed).

- $H_0 : \mu = 7.5$ vs. $H_A : \mu \neq 7.5$
- $n = 10, \bar{x} = 7.55, s = 0.1027$
- $SE = 0.1027/\sqrt{10} = 0.0325$
- $t^* = (7.55 - 7.5)/0.0325 = 1.54$
- P-value = 0.158

Critical Value Approach

Critical values: $\pm t_{0.025,9} = \pm 2.2616$; Since $-2.2616 < 1.54 < 2.2616 \rightarrow$ Fail to reject H_0

P-Value Approach

Since $0.158 > 0.05 \rightarrow$ Fail to reject H_0

Conclusion: Insufficient evidence that mean thickness differs from 7.5.

End of Unit II

Inferential Statistics & Hypothesis Testing

Next: Unit III — Basics of Python Programming

Hypothesis Testing for One-Sample Proportion

The same logic extends to testing population proportions.

Test Statistic

$$z^* = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

Conditions for Normal Approximation

- $np_0 \geq 5$
- $n(1 - p_0) \geq 5$

Example: Penn State Students from PA

Research question: Are the majority of Penn State students from Pennsylvania?

- $H_0 : p = 0.5$ vs. $H_A : p > 0.5$
- $n = 500$, 278 from PA $\rightarrow \hat{p} = 278/500 = 0.556$
- $\alpha = 0.05$

Check conditions:

- $np_0 = 500(0.5) = 250 \geq 5$ ✓
- $n(1 - p_0) = 250 \geq 5$ ✓

Test Statistic:

$$z^* = \frac{0.556 - 0.5}{\sqrt{\frac{0.5(0.5)}{500}}} = 2.504$$

Penn State Example: Decision

Critical Value Approach

Critical value $z_{0.05} = 1.645$

Since $2.504 > 1.645 \rightarrow \text{Reject } H_0$

P-Value Approach

P-value = $P(Z \geq 2.504) = 0.0062$

Since $0.0062 \leq 0.05 \rightarrow \text{Reject } H_0$

Conclusion: There is sufficient evidence that a majority of Penn State students are from Pennsylvania.

Connecting CI to Two-Tailed Tests

A **95% confidence interval** can be used to conduct a two-tailed test at $\alpha = 0.05$:

If the hypothesized value μ_0 (or p_0) **falls inside** the CI $\rightarrow$ **Fail to reject H_0**

If the hypothesized value μ_0 (or p_0) **falls outside** the CI $\rightarrow$ **Reject H_0**

Example: 95% CI for president approval is (0.415, 0.465). Testing $H_0 : p = 0.40$.

Since 0.40 is **not** in (0.415, 0.465) $\rightarrow$ **Reject H_0**

This connection applies to **all two-sided tests** and confidence intervals.

Summary: Critical Value vs. P-Value

Feature	Critical Value Approach	P-Value Approach
What you compare	Test statistic vs. critical value	P-value vs. α
Decision rule	Reject if $t \geq t^*$	$t \geq t^*$
Information needed	α and distribution table	α and software/table
Advantage	Clear cutoff	More informative (gives exact strength of evidence)

Both approaches always lead to the same conclusion!

Six Steps for Any Hypothesis Test

1. Set up hypotheses and check conditions
2. Decide on α (significance level)
3. Calculate the test statistic
4. Find rejection region (critical value) or calculate P-value
5. Make a decision about H_0
6. State an overall conclusion in context

Key Takeaways

- We assume H_0 is true and look for evidence against it
- We never "accept" H_0 — we only fail to reject it
- Type I Error (α): Rejecting a true H_0
- Type II Error (β): Failing to reject a false H_0
- The P-value tells you the strength of evidence against H_0
- Both the critical value and P-value approaches yield identical conclusions
- These concepts extend to all hypothesis tests you'll encounter

2.1 Types of Distributions

3 main types of sampling distributions are:

1. Sampling Distribution of Mean ($\bar{x}$) : Most common; used to estimate the population mean.
2. Sampling Distribution of Proportion ($\hat{p}$): Used for categorical data to estimate population proportions.
3. T-Distribution: When sampling from a normal population with unknown variance.

Sampling Distribution of Mean

The sampling distribution of the mean refers to the probability distribution of sample means that you get by repeatedly taking samples (of the same size) from a population and calculating the mean of each sample.

2.2 Key concepts of Sampling Distribution of Mean

Population Mean (μ): The average of the entire population.

Sample Mean ($\bar{x}$): The average of a sample taken from the population.

Sampling Distribution of the Mean: If you take multiple samples and plot their means, that plot will form the sampling distribution of the mean.

For any population with mean μ and standard deviation σ :

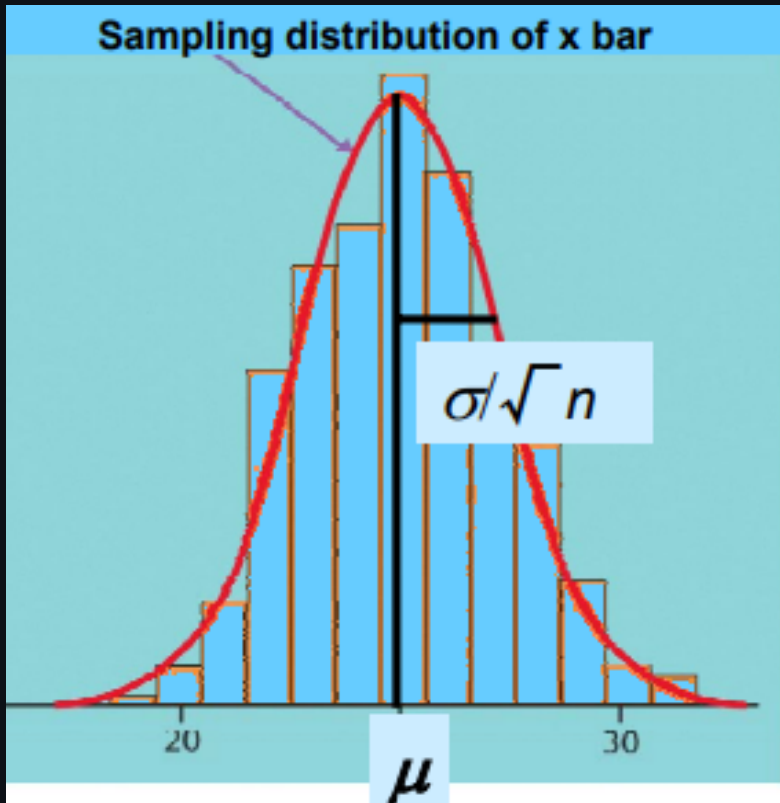
- Mean, or center of the sampling distribution of $\bar{x}$, is equal to the population mean, μ .

$$\mu_{\bar{x}} = \mu$$

- Standard deviation of the sampling distribution is $\sigma/\sqrt{n}$, where n is the sample size.

$$\sigma_{\bar{x}} = \sigma/\sqrt{n}$$

Source: <https://www.geeksforgeeks.org/maths/sampling-distribution/>



- $\sigma_{\bar{x}}$ - Standard Deviation of Sampling Deviation
- σ - Population Standard Deviation
- n - Sample size

Standard Error (SE)

This is just the practical name given to the standard deviation of a sampling distribution when you use it to measure the precision or accuracy of your sample estimate.

$$SE_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Where:

- **sigma** = Population standard deviation
- **n** = Sample size

As n increases, SE decreases → more precise estimates.

The Example Setup:

Imagine a large school where all students take a test.

- The true population mean score (μ) is 500 &
- The population standard deviation (σ) is 100.
- Instead of testing every student, we pick a random sample size (n) of 36 students and calculate their average score.

If we repeat this process over and over with many different groups of 36 students, the different sample averages will form a bell-shaped sampling distribution centered at 500.

Calculating the Standard Error

To find out how much these group averages typically bounce around the true average of 500,

- we compute the standard error using the formula: $(SE = \frac{\sigma}{\sqrt{n}})$.
- Population Standard Deviation (σ): 100 ; Sample Size (n): $\sqrt{36}$; Calculation: $(SE = \frac{100}{6} = 16.67)$

For a sample of 36 students with a mean of 515, the sampling distribution of the mean is centered at 500, and the standard error is $100 / \sqrt{36} = 16.67$

Key Takeaways

- Meaning: Our sample mean of 36 students will typically vary by about 16.67 points from the true school average of 500 just by random chance.
- Impact of Sample Size: If we make our sample bigger, say 400 students, the square root of 400 is 20, making the new $(SE = \frac{100}{20} = 5)$. A larger sample size shrinks the standard error, making our guess much more precise

The Example Setup (case b)

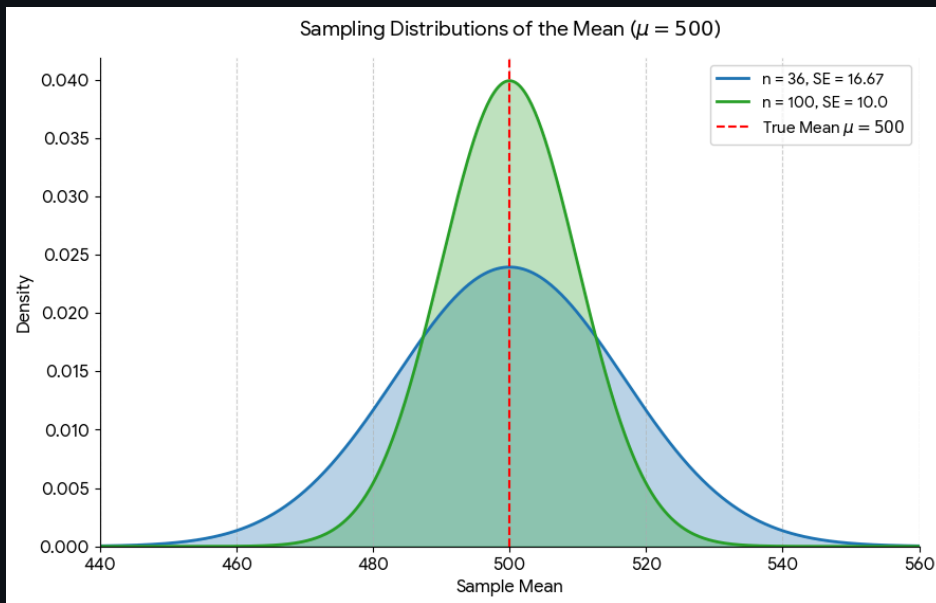
We will use the exact same large school from before:

- The true population mean score (μ) is 500.
- The population standard deviation (σ) is 100.
- This time, instead of 36 students, we increase our sample size (n) to 100 students.
- If we repeatedly select random groups of 100 students, the new sample averages will create a much narrower, tighter bell-shaped sampling distribution still centered at 500.

Calculating the New Standard Error

To see how much these larger group averages vary from the true average, we apply the formula: $(SE = \frac{\sigma}{\sqrt{n}})$.

- Population Standard Deviation (σ): 100
 - New Sample Size (n): 100
 - Square Root of Sample Size ($\sqrt{n}$): 10
- Calculation: $(SE = \frac{100}{10} = \mathbf{10})$



Sampling Distribution of the Sample Proportion

When dealing with categorical data, we often look at the proportion of successes in a sample.

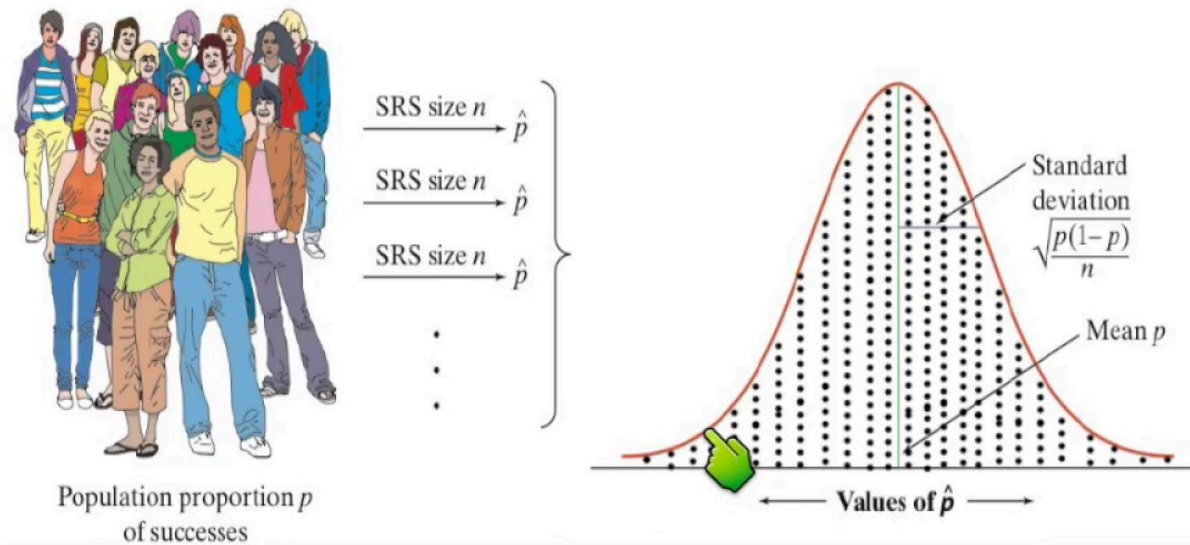
Key Features:

Mean: $\mu_{\hat{p}} = p$ (population proportion)

Standard error: $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$

This distribution is approximately normal if n is large and p is not too close to 0 or 1.

Sampling Distribution of a Sample Proportion (approximated to follow a Normal distribution)



Feature	Sample Mean ($\bar{x}$)	Sample Proportion ($\hat{p}$)	t-Distribution
Definition	The average of sample observations.	The ratio of 'successes' to total observations in a sample.	A probability distribution used to estimate population parameters when sample size is small and/or population variance is unknown.
Formula	$\bar{x} = \sum x_i / n$	$\hat{p} = \text{number of successes} / n$	Based on $t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$; $\text{CI} = \bar{x} \pm Z_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$
Type of Data	Quantitative (numerical)	Categorical (success/failure)	Quantitative; used with sample means
Parameter Estimated	Population mean (μ)	Population proportion (p)	Population mean (μ) when σ unknown
Distribution Assumption	Normal distribution if n is large (Central Limit Theorem)	Binomial distribution approximated by normal for large n	Approaches normal distribution as n increases
Use in Hypothesis Testing	Test population mean (μ) using t-test or z-test	Test population proportion using z-test	Used in t-tests for mean and confidence intervals (small samples)
Example	Average height of 30 students	Proportion of students who passed an exam	Estimating mean weight of a sample of 10 people when population variance is unknown

1. Sample Mean

Definition: The sample mean is the average of numbers in a sample drawn from a population.

Example: 10,20,30,40 ; $100/4=25$

2. Sample Proportion

Definition: The sample proportion is the fraction of observations in a sample that have a certain characteristic.

Example: survey of 20 students asks if they like mathematics. 14 say "Yes". $14/20=0.7$

3. t-Distribution (for Small Samples)

Definition: The t-distribution is used to estimate population means from small samples when the population standard deviation (σ) is unknown. It allows construction of confidence intervals that account for sample variability.

Example: Using the student scores above: [70, 75, 80, 85, 90] ; $sd=7.91$; $CI = \bar{x} \pm T_{\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$

T-value from the t-distribution; degrees of freedom ($df=n-1$) ; Significance level (α)

3.2 Z-Score

A z-score tells you how many standard deviations a data point is from the mean. It is calculated using the value, mean, and standard deviation.

Calculation Formula:

$$z = \frac{x - \mu}{\sigma}$$

x: The raw data point you want to check

μ: The mean (average) of the data set

σ: The standard deviation of the data set

Further Reading: <https://www.geeksforgeeks.org/data-science/z-test/>

Z Score Formula

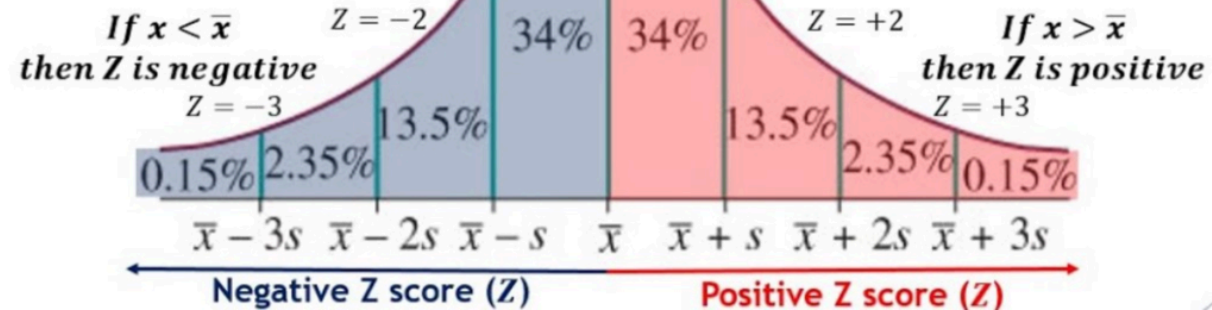


$$Z = \frac{X - \mu}{\sigma}$$



$\bar{x}$ = mean score

x = individual score



Z-Score Interpretation

Z-Score	Interpretation
$Z = 0$	Exactly at the mean
$Z = +1$	1 SD above mean (84th percentile)
$Z = -2$	2 SD below mean (2.5th percentile)
$Z = +2.5$	2.5 SD above mean (99.4th percentile)

Source: <https://www.investopedia.com/terms/z/zscore.asp>

	A	B	C	D
1	Factor (x)	Mean (μ)	St. Dev. (σ)	Z-Score
2	3	12.17	6.4	-1.43
3	13	12.17	6.4	0.13
4	8	12.17	6.4	-0.65
5	21	12.17	6.4	1.38
6	17	12.17	6.4	0.75
7	11	12.17	6.4	-0.18

How to Calculate Z-Score using Spreadsheets:

Cell D2 = (A2 - B2) / C2

Cell D3 = (A3 - B3) / C3

How to Read the Score

- Zero (0): The data point is the exact same as the mean.
- Positive number: The data point is above the mean. (Example: A score of +2 means it is two steps above the average).
- Negative number: The data point is below the mean. (Example: A score of -1 means it is one step below the average).

Why People Use It?

- Compare groups: It helps match and compare numbers from totally different types of tests or charts.
- Find unusual data: Scores higher than 3 or lower than -3 are very rare and are often treated as unusual outliers

End of Unit II

Inferential Statistics & Hypothesis Testing

Next: Unit III — Basics of Python Programming

Mathematical Form

If $X_1, X_2, \dots, X_n$ are i.i.d. with mean μ and variance σ^2 , then:

$$\bar{X} \xrightarrow{d} N\left(\mu, \frac{\sigma^2}{n}\right) \text{ as } n \rightarrow \infty$$

- Mean of the sample means:

$\mu_{\bar{X}} = \mu$: *The mean of the sample distribution equals the population mean μ .*

- Standard deviation of the sample mean (Standard Error):

$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$: *he population standard deviation σ divided by the square root of the sample size n .*

- Z-Score Formula for Probability:

$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1)$: *The population standard deviation σ divided by the square root of the sample size n .*

4.1 CLT — Why It Matters

Before CLT

- Limited to normally distributed populations
- Restricted analytical tools

After CLT

- **Universal applicability:** Works for ANY population distribution
- **Foundation of inference:** Enables confidence intervals & hypothesis testing
- **Sample size guideline:** $n \geq 30$ is the rule of thumb

Solved Problem

The Problem:

The weights of packages at a factory have a mean μ of 200 grams and a standard deviation σ of 15 grams.

If you take a random sample of ($n = 36$) packages, what is the probability that the sample mean weight $\bar{x}$ is less than 197 grams?

Step-by-Step Solution

1. Identify the Given Values

- Population mean, $\mu = 200$
- Population standard deviation, $\sigma = 15$
- Sample size, $n = 36$
- Target sample mean, $\bar{x} = 197$

2. Calculate the Standard Error of the Mean ($\sigma_{\bar{x}}$)

The formula for the standard error is: $(\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}})$

$$\sigma_{\bar{x}} = \frac{15}{\sqrt{36}} = \frac{15}{6} = 2.5$$

3. Compute the (Z)-Score

The formula to convert a sample mean to a standard normal (Z)-score is:

$$Z = \left(\frac{x - \mu}{\sigma_{\bar{x}}} \right) \quad Z = \frac{197 - 200}{2.5} = \frac{-3}{2.5} = -1.2$$

4. Find the Probability

Look up ($Z = -1.2$) in a standard normal distribution table. The area to the left of ($Z = -1.2$) is 0.1151.

Therefore, there is an 11.51 chance that a random sample of 36 packages will have a mean weight of less than 197 grams.

Solved Problem: Applying CLT

Problem: A factory produces light bulbs. The lifetime of bulbs is exponentially distributed with mean = 1000 hours and standard deviation = 1000 hours. A quality inspector takes samples of 50 bulbs.

Questions:

1. What is the shape of the population distribution?
2. What is the shape of the sampling distribution of sample means?
3. What is the mean of the sampling distribution?
4. What is the standard error?
5. What is the probability that a sample of 50 bulbs has average lifetime > 1100 hours?

Solution:

Part 1: Population is **exponentially distributed** (highly skewed right).

Part 2: By CLT, sampling distribution of means is **approximately normal** ($n = 50 > 30$).

Part 3: Mean of sampling distribution = population mean = **1000 hours**.

Part 4: Standard Error

$$SE = \sigma / \sqrt{n} = 1000 / \sqrt{50} = 1000 / 7.07 = 141.42 \text{ hours}$$

Part 5: Probability that sample mean > 1100

$$Z = \frac{\bar{X} - \mu}{\sigma / \sqrt{n}} \approx N(0, 1), \bar{X} = 1100, \mu = 1000, n = 50, \sigma = 1$$

$$Z = (1100 - 1000) / 141.42 = 100 / 141.42 = 0.707$$

From Z-table, $P(Z < 0.707) = 0.760$

So $P(Z > 0.707) = 1 - 0.760 = 0.240$

Answer: There is a 24% probability that a random sample of 50 bulbs will have average lifetime exceeding 1100 hours.

4.2 Central Limit Theorem - Demonstration

How the Concepts Connect: [Gamification]

Demo: <https://www.mathsisfun.com/data/quincunx.html>

- **Central Limit Theorem:** As the balls hit many pegs, each bounce is a random left or right choice. The final bin where a ball lands is the sum of these random steps, which physically proves that adding independent random events results in a bell-shaped pattern.
- **Normal Distribution:** The final piles of balls at the bottom of the board form a symmetric bell curve, which is a visual picture of a normal distribution.
- **Binomial Connection:** Each ball's path follows a binomial distribution, and the board shows how a binomial distribution turns into a normal distribution when there are many rows of pegs.

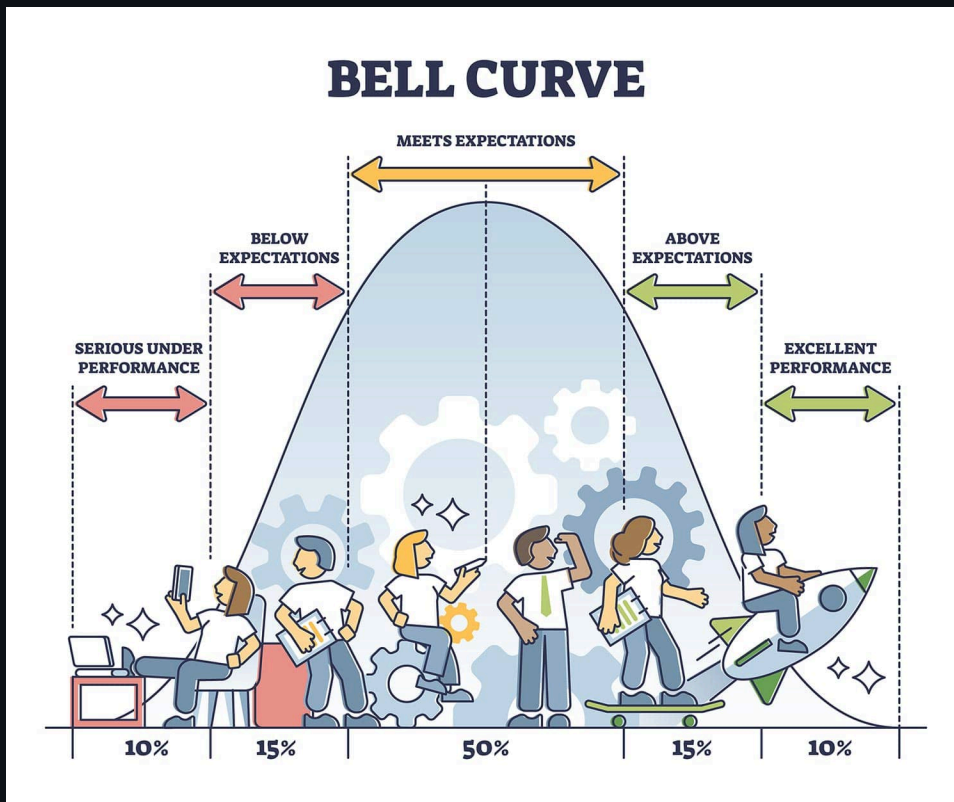
Explanation: Reference: <https://www.mathsisfun.com/data/quincunx-explained.html>

4.3 Core Reasons for Importance of Normal Distribution

The normal distribution is vital in statistics because it drives the Central Limit Theorem, models real-world data like human heights and test scores, and simplifies hypothesis testing and confidence intervals.

- **Central Limit Theorem:** Averages of random samples form a normal curve as sample size grows, even if original data is not normal. [Example in next slide]
- **Real-World Fit:** Many natural and social traits (weight, blood pressure, IQ) naturally cluster in a bell shape around an average.
- **Easy to Measure:** Defined completely by just two values: the mean (center) and standard deviation (spread).
- **The Empirical Rule:** Roughly 68% of data falls within 1 standard deviation of the mean, 95% within 2, and 99.7% within 3.
- **Statistical Inference:** Powers standard tests and confidence limits used in research and data analysis.

- Averages of random samples form a normal curve as sample size grows, even if original data is not normal.
- If you take repeated groups of numbers from a non-normal dataset, the averages of those groups will always form a bell curve, provided the groups are large enough.



Normal Distribution Example

4.4 Application of CLT

Example/ Demonstration: The "Human Height" Quick-Sample Run

This leverages your students' own physical data to prove that small, random groups average out predictably.

- The Setup: This requires a class of at least 20–30 students.

[Activity]:

1. Ask everyone to stand up.
2. Designate one student as the "Randomizer." They will blindly point to 5 random students in the room.
3. Those 5 students must quickly average their heights (or birth months, to keep it simpler/less sensitive) and shout it out.
4. Plot that single average on the board.
5. Repeat this 10 times with different random groups of 5.

The Reveal: Even if one group accidentally contains the tallest three people in the room, the other groups will naturally balance out. The plotted points will instantly cluster heavily around the true classroom average, perfectly demonstrating the stabilizing power of the CLT.

4.5 Real-World Example: Exam Scores

Suppose:

- Mean (μ) = 70
- Standard Deviation (σ) = 10

Score Range	Percentage of Students
60 – 80	68%
50 – 90	95%
40 – 100	99.7%

Interpretation:

- Most students score between 60 and 80.
- Very few score below 40 or above 100.

4.6 Normal Distribution vs Skewed Distribution

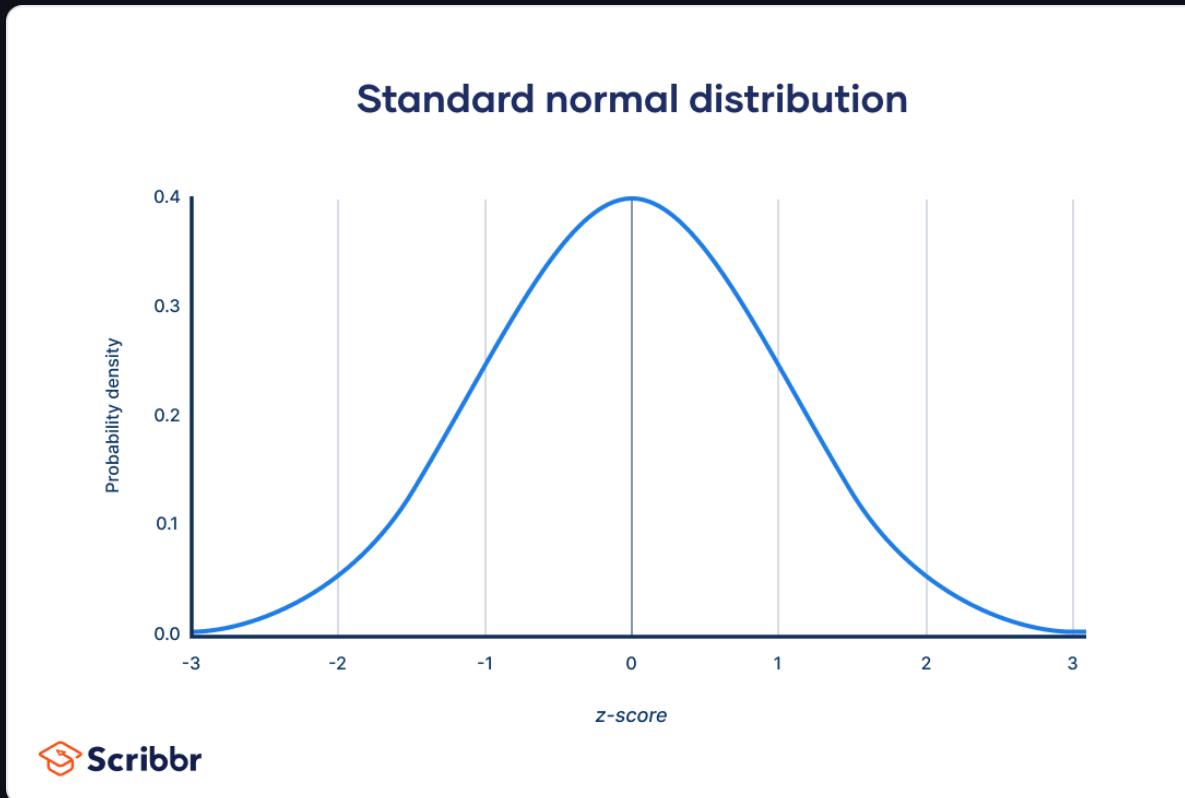
Normal Distribution	Skewed Distribution
Symmetric	Asymmetric
Mean = Median = Mode	Mean $\neq$ Median $\neq$ Mode
Bell-shaped	Tail extends to one side
No skewness	Positive or Negative skewness

4.7 Quick Summary of Normal Distribution

Feature	Normal Distribution
Shape	Bell-shaped
Symmetry	Symmetric
Mean = Median = Mode	Yes
Center	Mean (μ)
Spread	Standard Deviation (σ)
Total Area	100% (or 1)
Common Rule	68–95–99.7 Rule

5 Standard Normal Distribution (Z-Distribution)

The standard normal distribution, or z-distribution, is a special bell-shaped probability curve with a mean (μ) of 0 and a standard deviation (σ) of 1. Individual values are converted into z-scores to measure distance from the mean in standard deviation units.



5.1 (Z-Distribution) Key Characteristics

- **Center and Spread:** The mean, median, and mode are all equal to 0, and the spread equals 1.
- **Symmetry:** The curve is perfectly symmetric around 0.
- **Total Area:** The total area under the curve equals 1 (or 100%).

A Z distribution table, also known as a Z-score table, provides the probabilities associated with Z-scores in a standard normal distribution, helping you find the area under the curve for specific values.

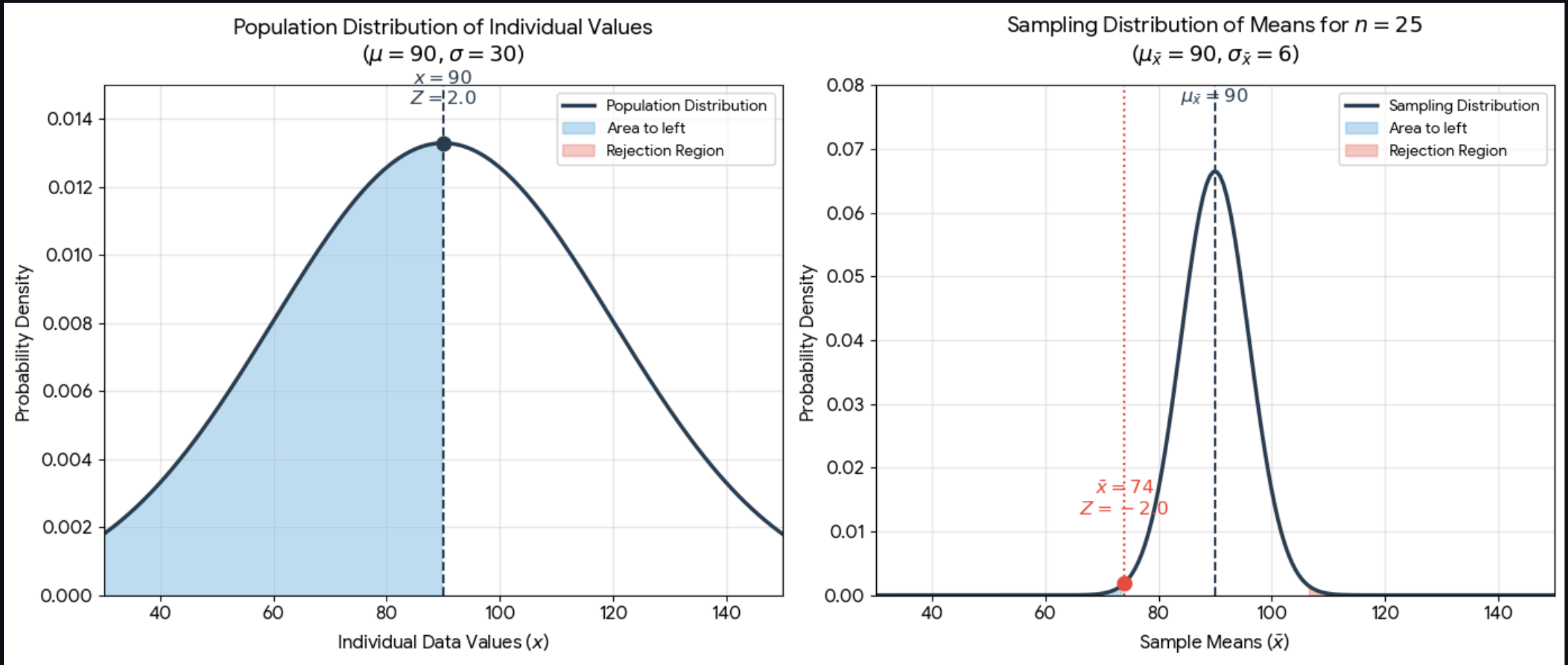
5.2 Key Differences At a Glance

Aspect	Z-Score	Z-Test
Purpose	Measures position of a single value	Tests hypothesis about a population mean
Use	Descriptive statistic	Inferential statistic
Formula	$(Z = \frac{x - \mu}{\sigma})$	$(Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}})$
Denominator	Population standard deviation	Standard error (adjusted for sample size)
Data Type	Single value	Sample mean

Key point

Z-score: Measures how many standard deviations a single value is from the mean.

Z-test: Tests if a sample mean significantly differs from the population mean using the standard error.



Practical: Normal Distribution in Python

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

# Parameters
mu, sigma = 100, 15 # IQ distribution

# Generate data
x = np.linspace(mu - 4*sigma, mu + 4*sigma, 1000)
y = stats.norm.pdf(x, mu, sigma)

# Plot
plt.figure(figsize=(10, 6))
plt.plot(x, y, 'b-', linewidth=2, label=f'N({mu}, {sigma}^2)')
plt.fill_between(x, y, where=(x >= mu-sigma) & (x <= mu+sigma),
                 alpha=0.3, label='68%')
plt.fill_between(x, y, where=(x >= mu-2*sigma) & (x <= mu+2*sigma),
                 alpha=0.2, label='95%')
plt.axvline(mu, color='red', linestyle='--', label=f'Mean = {mu}')
plt.title('Normal Distribution: IQ Scores')
plt.xlabel('IQ Score'); plt.ylabel('Probability Density')
plt.legend(); plt.grid(True, alpha=0.3)
plt.show()
```

Python Z-Score Calculation

```
# Business Example: Employee performance scores
scores = np.array([72, 85, 90, 68, 95, 78, 88, 82, 91, 75])
mu_scores = np.mean(scores)
sigma_scores = np.std(scores, ddof=1)

# Z-score for a score of 90
x = 90
z = (x - mu_scores) / sigma_scores
print(f"Z-score for {x}: {z:.3f}")

# Percentile
percentile = stats.norm.cdf(z) * 100
print(f"Percentile: {percentile:.1f}%")

# Probability: P(X > 85)
z_85 = (85 - mu_scores) / sigma_scores
prob_greater = 1 - stats.norm.cdf(z_85)
print(f"P(X > 85) = {prob_greater:.4f}")
```

CLT Demonstration in Python

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(42)
# Population: Exponential distribution (highly skewed)
population = np.random.exponential(scale=2, size=100000)
sample_sizes = [1, 5, 30, 100] # Sample sizes to demonstrate
fig, axes = plt.subplots(2, 2, figsize=(12, 10))
for idx, n in enumerate(sample_sizes):
    ax = axes[idx // 2, idx % 2]
    sample_means = [np.mean(np.random.choice(population, n)) # Take 10,000 samples of size n, compute means
                    for _ in range(10000)]
    ax.hist(sample_means, bins=50, density=True, alpha=0.7,
            color='steelblue', edgecolor='black')
    ax.set_title(f'Sample Size n = {n}')
    ax.set_xlabel('Sample Mean'); ax.set_ylabel('Density')
    if n >= 30: # Overlay normal curve for n >= 30
        mu, se = np.mean(sample_means), np.std(sample_means)
        x = np.linspace(mu - 4*se, mu + 4*se, 100)
        ax.plot(x, stats.norm.pdf(x, mu, se), 'r-', lw=2, label='Normal fit')
    ax.legend()
plt.suptitle('Central Limit Theorem Demonstration\n(Exponential Population)', fontsize=14)
plt.tight_layout()
plt.show()
```

6. Hypothesis Testing — Framework

Definition

Hypothesis Testing is a statistical method used to make decisions about a population parameter based on sample evidence. It helps us determine whether observed differences or relationships are statistically significant or just due to random chance.

The Two Hypotheses

Hypothesis	Symbol	Statement	Role
Null Hypothesis	H_0	Status quo, no effect, no difference	Assumed true initially
Alternative Hypothesis	H_1 or H_a	Claim to be tested	What we seek evidence for

Further Reading: <https://www.geeksforgeeks.org/data-science/understanding-hypothesis-testing/>

6.1 Key Terms of Hypothesis Testing

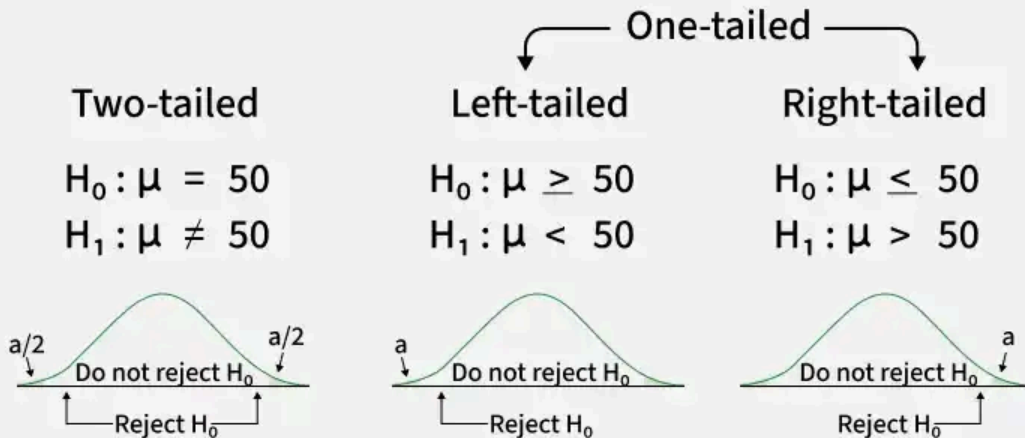
- Significance Level (α): How sure we want to be before saying the claim is false. Usually, we choose 0.05 (5%).
- p-value: The chance of seeing the data if the null hypothesis is true. If this is less than α , we say the claim is probably false.
- Test Statistic: A number that helps us decide if the data supports or rejects the claim.
- Critical Value: The cutoff point to compare with the test statistic.
- Degrees of freedom: A number that depends on the data size and helps find the critical value.

6.2 Types of Tests

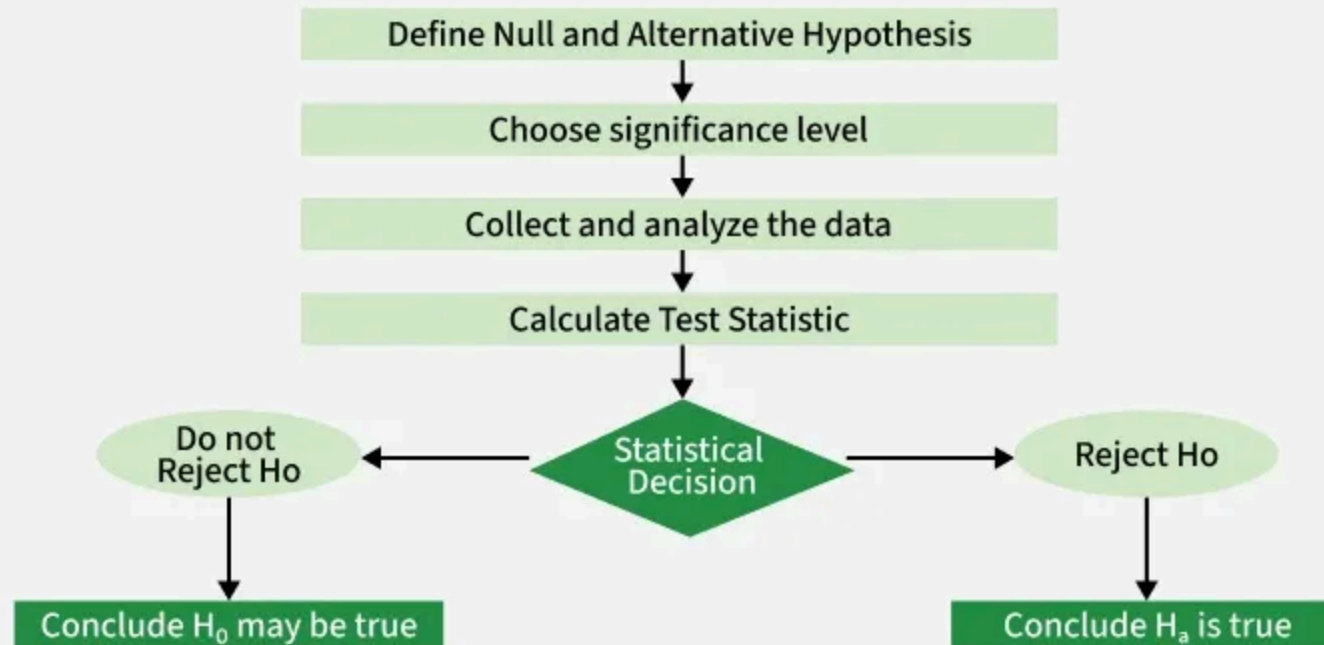
By Directionality

Two-tailed test: $H_1: \mu \neq \mu_0 \rightarrow$ Detect any difference
Right-tailed: $H_1: \mu > \mu_0 \rightarrow$ Detect increase
Left-tailed: $H_1: \mu < \mu_0 \rightarrow$ Detect decrease

Hypothesis Testing



6.3 How does Hypothesis Testing work



Calculate Test Statistic

The test statistic measures how much the sample data deviates from what we did expect if the null hypothesis were true. Different tests use different statistics:

- Z-test: Used when population variance is known and sample size is large [≥ 30].
- T-test: Used when sample size is small or population variance unknown [< 30].
- Chi-square test: Used for categorical data to compare observed vs. expected counts.

Make a Decision

We compare the test statistic to a critical value from a statistical table or use the p-value:

1. Using Critical Value:

- If test statistic $>$ critical value $\rightarrow$ reject H_0 .
- If test statistic $\leq$ critical value $\rightarrow$ fail to reject H_0 .

2. Using P-value:

- If p-value $\leq \alpha \rightarrow$ reject H_0 .
- If p-value $> \alpha \rightarrow$ insufficient evidence to reject H_0 , not proof that H_0 is true.

6.2 Test Statistic & Rejection Region

Test Statistic (Z-Test)

$$Z = \frac{\bar{X} - \mu_0}{\sigma / \sqrt{n}}$$

Decision Rule

If $|Z| > Z_{\text{critical}}$ → Reject H_0
If $|Z| \leq Z_{\text{critical}}$ → Fail to reject H_0

Critical Values (Common)

Confidence Level	α	Two-tailed $Z_{\alpha/2}$	One-tailed Z_{α}
90%	0.10	± 1.645	1.282
95%	0.05	± 1.96	1.645
99%	0.01	± 2.576	2.326

5.3 P-Value Approach

Definition

The **P-value** is the probability of obtaining a test statistic at least as extreme as the one observed, assuming H_0 is true.

Decision Rule

If $p\text{-value} \leq \alpha \rightarrow$ Reject H_0 (Result is statistically significant)
If $p\text{-value} > \alpha \rightarrow$ Fail to reject H_0

P-Value Interpretation

P-Value	Evidence Against H_0
$p > 0.10$	Weak / No evidence
$0.05 < p \leq 0.10$	Moderate evidence
$0.01 < p \leq 0.05$	Strong evidence
$p \leq 0.01$	Very strong evidence

5.4 Type I and Type II Errors

Error Matrix

	H_0 is True	H_0 is False
Reject H_0	Type I Error (α)	✓ Correct (Power)
Fail to Reject H_0	✓ Correct	Type II Error (β)

Definitions

- α (Significance level): $P(\text{Type I error}) = P(\text{Reject } H_0 \mid H_0 \text{ true})$
- β : $P(\text{Type II error}) = P(\text{Fail to reject } H_0 \mid H_0 \text{ false})$
- **Power** = $1 - \beta = P(\text{Reject } H_0 \mid H_0 \text{ false})$

Business Analogy

Error	Analogy	Consequence
Type I	False alarm	Launching a bad product
Type II	Missed opportunity	Rejecting a good product

Error Matrix

	H_0 is True	H_0 is False
Reject H_0	Type I Error (α)	✓ Correct (Power)
Fail to Reject H_0	✓ Correct	Type II Error (β)

5.5 Practical: One-Sample Z-Test

```
import numpy as np
from scipy import stats

# Business Problem:
# A company claims average delivery time is 30 minutes.
# Sample of 40 deliveries: mean = 32.5 min, known  $\sigma = 5$  min
# Test at  $\alpha = 0.05$ 

mu_0 = 30      # Hypothesized mean
x_bar = 32.5    # Sample mean
sigma = 5      # Known population SD
n = 40         # Sample size
alpha = 0.05

# Z-statistic
z_stat = (x_bar - mu_0) / (sigma / np.sqrt(n))
print(f"Z-statistic: {z_stat:.4f}")

# Critical value (two-tailed)
z_critical = stats.norm.ppf(1 - alpha/2)
print(f"Critical value ( $\pm$ ): {z_critical:.4f}")

# P-value
p_value = 2 * (1 - stats.norm.cdf(abs(z_stat)))
print(f"P-value: {p_value:.6f}")

# Decision
if p_value < alpha:
    print("Result: REJECT  $H_0$  – Delivery time significantly differs from 30 min")
else:
    print("Result: FAIL TO REJECT  $H_0$  – No significant difference")
```

5.6 Practical: Two-Sample Z-Test (A/B Testing)

```
# Business Problem: A/B Testing
# Group A (old website):  $n_1=100$ ,  $\bar{x}_1=120$ ,  $\sigma_1=15$ 
# Group B (new website):  $n_2=100$ ,  $\bar{x}_2=128$ ,  $\sigma_2=15$ 
# Test if new website has higher conversion at  $\alpha=0.05$ 

n1, n2 = 100, 100
x1, x2 = 120, 128
s1, s2 = 15, 15
alpha = 0.05

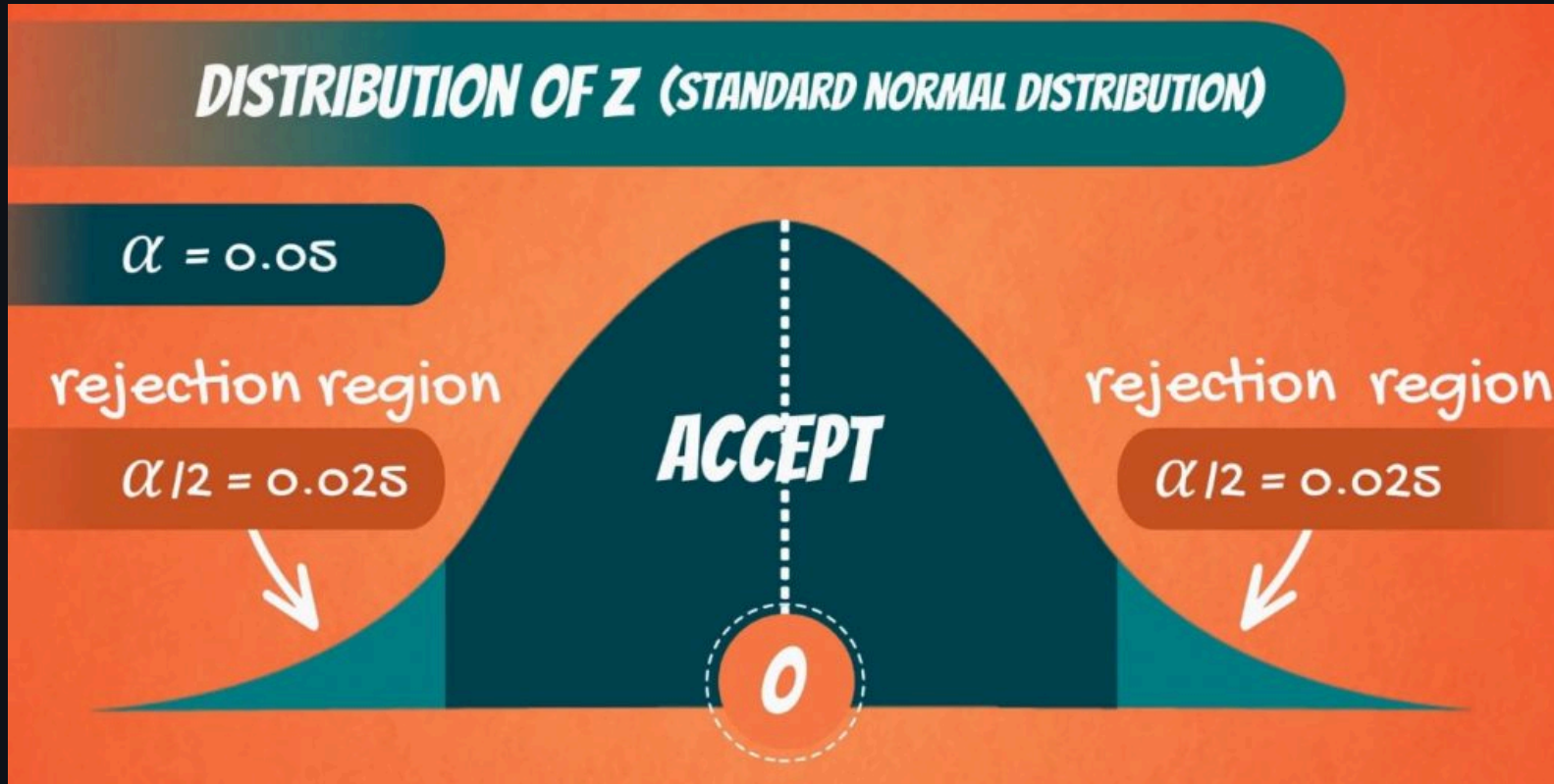
# Pooled standard error
se = np.sqrt((s1**2 / n1) + (s2**2 / n2))

# Z-statistic
z_stat = (x2 - x1) / se
print(f"Z-statistic: {z_stat:.4f}")

# P-value (one-tailed: right)
p_value = 1 - stats.norm.cdf(z_stat)
print(f"P-value: {p_value:.6f}")

if p_value < alpha:
    print("✅ New website significantly outperforms old ( $p < 0.05$ )")
else:
    print("❌ No significant difference detected")
```

Rejection Region



5.7 Confidence Intervals [content covered beyond syllabus]

Definition

A **confidence interval** gives a range of plausible values for an unknown population parameter.

Formula (Known σ)

$$CI = \bar{X} \pm Z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

Interpretation

"We are 95% confident that the true population mean lies within this interval."

Relationship with Hypothesis Testing

If μ_0 falls INSIDE the 95% CI $\rightarrow$ Fail to reject H_0 at $\alpha=0.05$
If μ_0 falls OUTSIDE the 95% CI $\rightarrow$ Reject H_0 at $\alpha=0.05$

5.8 Practical: Confidence Intervals

```
# 95% Confidence Interval for delivery time example
x_bar = 32.5
sigma = 5
n = 40
confidence = 0.95

z_alpha = stats.norm.ppf((1 + confidence) / 2)
margin_error = z_alpha * (sigma / np.sqrt(n))

ci_lower = x_bar - margin_error
ci_upper = x_bar + margin_error

print(f"95% Confidence Interval: [{ci_lower:.3f}, {ci_upper:.3f}]")
print(f"Margin of Error: ±{margin_error:.3f}")

# Interpretation
print(f"\nWe are 95% confident the true mean delivery time")
print(f"is between {ci_lower:.2f} and {ci_upper:.2f} minutes.")
```

6. Summary & Key Takeaways

Concept	Key Point
Normal Distribution	$X \sim N(\mu, \sigma^2)$, symmetric, bell-shaped
Z-Score	Standardization: $Z = \frac{X-\mu}{\sigma}$
CLT	$\bar{X} \approx N(\mu, \sigma^2/n)$ for large n
H_0 vs H_1	Null = status quo; Alternative = claim
Test Statistic	$Z = \frac{\bar{X}-\mu_0}{\sigma/\sqrt{n}}$
P-value	P(data H_0 true); reject if $p \leq \alpha$
Type I Error	False positive; controlled by α
Type II Error	False negative; reduced by increasing n

Decision Flowchart

Define H_0 and H_1 → Choose α → Collect sample
→ Compute test statistic → Find p-value
→ Compare p-value with α → Make decision

Exercises

1. **Z-Score:** IQ score = 130, $\mu = 100$, $\sigma = 15$. Find percentile.
2. **CLT:** Explain why sample mean of 50 dice rolls approximates normal.
3. **Hypothesis Test:** $H_0 : \mu = 50$, $\bar{x} = 53$, $\sigma = 10$, $n = 100$, $\alpha = 0.05$. Decision?
4. **P-value:** If $p = 0.03$ and $\alpha = 0.05$, what is your conclusion?
5. **Confidence Interval:** Compute 99% CI for problem 3.

References

1. Prasad, R.N. & Acharya, S. — *Fundamentals of Business Analytics* (Wiley)
2. Albright, S.C. & Winston, W.L. — *Business Analytics: Data Analysis & Decision Making* (Cengage)
3. Milton, M. — *Head First Data Analysis* (O'Reilly)

End of Unit II

Inferential Statistics & Hypothesis Testing

Next: Unit III — Basics of Python Programming

Probability Density Function (PDF) of the Normal Distribution (Gaussian Distribution).

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

Parameters

Symbol	Meaning	Example
$(f(x))$	Probability density at value (x)	Density at 70 marks
(x)	The value of the random variable	Marks = 75
(μ)	Mean (average)	70
(σ)	Standard deviation	10
(e)	Euler's number (≈ 2.718)	Mathematical constant
(π)	Pi (≈ 3.1416)	Mathematical constant

It is used to describe how data is distributed around the mean and to calculate the probability that a value falls within a certain range

Example

- Imagine the marks of 10,000 students.
- Most students score near the average.
- Very few score extremely low or extremely high.
- The formula calculates how high the bell-shaped curve is at any point x .

Probability Density



Real Example

Suppose

- Mean marks = 70
- Standard deviation = 10
- Question: What is the probability density at 70 marks?

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; \text{ Given: } x=70, \mu=70, \sigma=10$$

$$\text{Step 1: } x - \mu = 70 - 70 = 0$$

$$\text{Step 2: } (x - \mu / \sigma)^2 = (0 / 10)^2 = 0$$

$$\text{Step 3: } e^{-0} = 1$$

$$\text{Step 4: Substitute into the formula } f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$$f(70) = \frac{1}{10\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{70-70}{\sigma}\right)^2} \Rightarrow f(70) \approx 0.0399$$

- This is the maximum height of the PDF, because 70 is the mean.
- This is the probability density, not the probability.

For a continuous variable: $P(X=70)=0$

- To find an actual probability, you calculate the area under the PDF over an interval.

For example: $P(60 < X < 80) \approx 0.68$

- This means 68% of students score between 60 and 80.

Easy Analogy

Imagine a mountain.

- The peak of the mountain is the mean (μ).
- The width of the mountain depends on the standard deviation (σ).
- The formula tells you how high the mountain is at any location x .

It does not directly give probability. To obtain a probability, you measure the area under the mountain (curve) between two values.

1. PDF Formula (Height of the Curve): The PDF gives the height of the curve at a point.

This does not give probability. Think of it as measuring the height of a hill at one point.

2. Probability (Area Under the Curve)

To calculate the probability between two values a and b, we integrate the PDF.

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Substituting the normal PDF:

$$P(a \leq X \leq b) = \int_a^b f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx$$

This integral represents the area under the normal curve from a to b.

Why Don't We Solve It by Hand?

Unlike many simple functions, the normal distribution integral has no elementary closed-form solution.

(Mean the exact area under the curve cannot be expressed using a finite combination of basic, well-known functions like polynomials, exponentials, logarithms, and trigonometric functions.)

That's why mathematicians use:

- Z-tables
- Scientific calculators
- Excel
- Python
- R
- Statistical software

Exam Tip

Students often confuse PDF with probability:

- PDF formula → Gives the density (height) of the normal curve at a point.
- Probability → Obtained from the area under the PDF curve between two values.

This is why we use the PDF to answer questions like

- "What is the probability that a student's marks are between 60 and 80?"
rather than
- "What is the probability that a student's marks are exactly 70?"

1.1 Properties of Normal Distribution

Property	Description
Symmetry	Symmetric about the mean μ
Mean = Median = Mode	All three coincide at μ
Asymptotic	Tails extend to $\pm\infty$ but never touch x-axis
Area under curve	Total area = 1 (100% probability)
Inflection points	At $\mu \pm \sigma$

Empirical Rule (68-95-99.7)

68% $\rightarrow$ $[\mu - \sigma, \mu + \sigma]$
95% $\rightarrow$ $[\mu - 2\sigma, \mu + 2\sigma]$
99.7% $\rightarrow$ $[\mu - 3\sigma, \mu + 3\sigma]$

Difference Between PMF and PDF (Topic beyond syllabus)

PMF (Probability Mass Function)	PDF (Probability Density Function)
Used for Discrete variables	Used for Continuous variables
Gives probability of an exact value	Gives probability over an interval
Example: Number of defective bulbs	Example: Height, Weight, Temperature
Probabilities add up to 1	Total area under the curve equals 1

5 Probability Density Function (PDF)

A Probability Density Function (PDF) describes the probability distribution of a continuous random variable.

- Instead of giving the probability at a single value, the PDF gives the density of probability.
- The probability is found by calculating the area under the curve over an interval.

Key Properties of PDF

- Applicable to continuous random variables.
- The PDF curve is always non-negative: $f(x) \geq 0$
- The total area under the PDF curve is 1 (100% probability).
- The probability of a value lying between a and b equals the area under the curve between a and b.

A continuous random variable is a variable that can take on an infinite, uncountable number of values within a given range

Real-World Example

A company manufactures bolts.

- Average length = 10 cm
- Standard deviation = 0.2 cm

The PDF shows how likely different bolt lengths are.

Bolt Length	Probability Density
9.4 cm	Low
9.8 cm	Medium
10.0 cm	Highest
10.2 cm	Medium
10.6 cm	Low

Interpretation: Most bolts are produced close to 10 cm, while very short or very long bolts are rare.